Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

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Contents

Abstract

Background. Treatment response in aggregated N-of-1 clinical trials reflects three causally distinct mechanisms: pharmacological action of the active compound (BR), the patient's belief that the treatment will work (PB), and natural-history drift in the underlying condition (TV). Standard analysis models lump these into a single 'treatment effect' plus residual error. Whether this lumping biases inference is not unconditional: it depends on the estimand and on whether candidate biomarkers correlate with the non-pharmacological components, a dependence this paper makes precise.

Methods. We formalise a three-component decomposition of the within-participant outcome trajectory, each component a modified Gompertz function with participant-specific random effects, and characterise the trial-design contrasts (open-label, blinded discontinuation, crossover) that supply identifiability. We derive the omitted-variable-bias identity for the one-component estimator, which expresses both the lumped treatment effect and the lumped biomarker slope as sums of component-specific terms, and we evaluate the analysis strategies in a pre-registered Monte Carlo study (Study A) calibrated to the prazosin-PTSD reference parameters, comparing a one-component design-contrast analysis with a phase-augmented analysis at 1,000 replicates per alternative cell (5,000 per null cell) across $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 100, 150\}$.

Results. The identity shows that the lumped biomarker-by-treatment slope is displaced from the pharmacological slope by a term proportional to the biomarker's covariance with PB and TV, not by the magnitude of those components. Study A confirms this: with a biomarker coupled to BR only, the one-component estimate

of the biomarker-by-treatment interaction is essentially unbiased across the whole grid (mean absolute bias 0.016 at $N = 35$ and 0.006 at $N = 150$, within one Monte Carlo standard error of zero), with near-nominal coverage and type I error. The phase-augmented analysis provides no bias reduction and is markedly less powerful (mean power 0.35 versus 0.59 at $N = 35$; 0.62 versus 0.90 at $N = 70$), so it is dominated under this uncontaminated biomarker. A contaminated-biomarker pilot then exhibits the failure mode directly: coupling the biomarker to PB drives the one-component bias up with the coupling strength, reaching +0.51 at a biomarker-PB correlation of 0.45 ($N = 150$, some 30 Monte Carlo standard errors from zero), and the operative quantity is the biomarker's correlation with the individual-level PB *variance*, not the population-mean placebo response, which does not enter the slope; the phase-augmented analysis does not remove this bias either. The inferential value of decomposition is therefore conditional, not universal.

Conclusions. The trial design must supply the drug-on/off and belief contrasts; given those, component-level decomposition is useful when the estimand is the attribution of response to pharmacology versus belief versus natural history, or when a candidate biomarker may correlate with PB or TV. For a biomarker-by-treatment interaction with a biomarker orthogonal to PB and TV, a single-indicator analysis that exploits the design contrast is sufficient and a parametric component model is counterproductive. A pragmatic specification for the cases that warrant it is `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase` with random intercept and AR(1) residual correlation. A contaminated-biomarker study demonstrates the failure mode (the lumped slope is biased in proportion to the biomarker's correlation with PB), and Study B shows that the bias is removable but only under a design that decouples drug exposure from belief: on a balanced-placebo design the belief-covariate decomposition recovers the unbiased pharmacological slope (bias within one Monte Carlo SE of zero across the feasible contamination range at N=150) while the one-component estimator does not, whereas on the standard hybrid design no decomposition recovers it. Recovery is therefore conditional on the trial design, not on the analysis alone.

1 Introduction

1.1 A claim about how N-of-1 trials should be analysed

[1]

> **This paper argues that treatment response in N-of-1 trials comprises three causally distinct components that must be analysed jointly rather than lumped.**
>
> - The three components are BR (pharmacological), PB (placebo-belief), and TV (natural history).
> - Each component is causally distinct, clinically interpretable, and identifiable from a well-designed N-of-1 trial.

- The standard single-indicator mixed-effects model conflates all three and renders biomarker validation unanswerable.

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This paper advances an opinionated methodological claim: treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components and should routinely be analysed as such, rather than estimated as a single lumped quantity. The three components are pharmacological action (BR, the biological response), placebo-belief response (PB, the patient's belief that they are receiving treatment), and time-variant natural history (TV, the trajectory that would have occurred without intervention). Each is causally distinct, each is independently of clinical and regulatory interest, and each is identifiable from a properly designed N-of-1 trial. We shall argue that the standard practice of fitting a single mixed-effects model with one treatment indicator, prevalent across the published N-of-1 literature, conflates these three components, biases the interpretation of treatment effect, and renders predictive-biomarker validation unanswerable as a scientific question.

[2]

The contribution is the consolidation and extension of an established structural-decomposition tradition, not a novel decomposition.

- The additive disease-progression, placebo, and drug-action decomposition is well established in pharmacometrics and longitudinal trial methodology.
- The present framework extends that tradition with an explicit, separately identified placebo-belief channel and brings it to the aggregated N-of-1 setting.
- The decomposition logic is not specific to N-of-1; it applies wherever a longitudinal treatment response is analysed.

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We make no claim of novelty for the decomposition itself. The structural decomposition of a longitudinal treatment response into separable disease-progression, placebo, and drug-action components is a well-established tradition in pharmacometrics[1–3], and the separation of drug from placebo response in longitudinal trajectories has been pursued both through growth-mixture models[4] and through experimental decomposition of the placebo response itself[5]. Each ingredient we use (mixed-effects models with trial-phase indicators, placebo-arm contrasts, regression-to-mean adjustment, growth-curve parameterisation) has likewise appeared across decades of clinical-trial methodology. Our contribution is therefore one of consolidation and extension rather than invention: we make the three-component structure explicit for the aggregated N-of-1 setting, add a separately modelled placebo-belief channel identified by design contrasts rather than by functional form alone, and argue that the decomposition should become routine wherever a longitudinal treatment response is analysed. We document why the framing has not been routine in N-of-1 practice (parsimony, identifiability concerns, sample-size constraints, methodological inertia) and propose an analysis that is tractable at trial-relevant sample sizes through a phase-augmented linear-mixed-effects approximation to the full decomposition.

1.2 Why this matters: three failure modes of the lumped analysis

[3]

The lumped one-component analysis has three documented failure modes that grow with unmodelled component heterogeneity.

- The three failure modes are placebo-attribution bias, regression-to-the-mean, and biomarker-validation failure.
- Each failure mode is documented in the chronic-condition trial literature, not merely hypothetical.
- The failures are present regardless of sample size.

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We begin with the failure modes of the standard approach. A one-component analysis ('the treatment effect is what we observe minus baseline') has three identifiable failure modes that arise when the population displays meaningful heterogeneity in any of the unmodelled components. None of these is hypothetical: each has a documented presence in the chronic-condition trial literature. Two points of precision, established formally in section 6.1 and borne out by the simulation of section 7.3, should be carried through what follows. The first two failure modes attach to the baseline-anchored lumped effect and are removed once the design supplies a drug-on/off contrast, which the one-component analysis of later sections in fact uses. The third, biomarker-validation failure, is different in kind: it requires not merely that PB or TV be large but that the candidate biomarker correlate with them, and where that coupling is absent the lumped analysis is not biased at all.

[4]

Placebo-attribution bias is sample-size-invariant and docu-

mented across antidepressant, anxiety, and IBS literatures.

- Lumped treatment-effect estimates overstate pharmacological efficacy in proportion to PB magnitude and variance.
- Kirsch et al. and Locher et al. demonstrated that antidepressant apparent efficacy is largely explained by placebo response and natural fluctuation.
- Placebo responsiveness has measurable biological correlates (e.g. COMT genotype), confirming it is a structured phenomenon, not noise.

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The first is the placebo-attribution failure. When the population contains both high and low placebo responders, the lumped treatment effect overestimates the pharmacological component. The bias scales with the magnitude and variance of PB in the study population, and it is invariant under sample-size increases: a larger one-component trial converges on the same biased estimate. The most consequential example in modern clinical pharmacology is antidepressant efficacy.[6] analysed FDA-submitted data on four widely- prescribed antidepressants and showed that, except at the most severe baseline-severity stratum, mean drug-placebo differences fell below clinical-significance thresholds, with the apparent 'efficacy' largely explained by placebo response and natural fluctuation;[7] reproduced this pattern in late-life depression and explicitly invoked regression to the mean.[8] generalised the finding to anxiety. Patient populations differ systematically in placebo responsiveness, and the difference has measurable biological correlates: COMT val158met genotype predicts placebo response in irritable bowel syndrome[9,10], indicating that 'placebo' is not a homogeneous nuisance but a structured biological phenomenon that interacts with the patient's genome and treatment history. A trial whose population happens to be placebo-responsive will report a larger 'drug effect' than one whose population is not, even when the underlying pharmacology is identical.

[5]

Regression-to-the-mean contributes several points of apparent improvement in early trial weeks and is absorbed without trace by the lumped analysis.

- Enrolment at peak severity guarantees a natural-history regression component irrespective of pharmacology.
- Hrobjartsson et al. systematic reviews across 200+ trials showed much of conventional placebo response was indistinguishable from natural-history drift.

- The lumped analysis has no formal mechanism to recover the pharmacological effect once regression is absorbed.

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The second failure mode is regression-to-the-mean. When patients enrol at moments of clinical engagement that coincide with peak symptom severity, natural-history regression toward typical severity contributes several points of apparent improvement in the early weeks of any trial, without any pharmacological contribution at all. The systematic reviews of[11],[12], and[13], the last covering 202 trials across 60 clinical conditions, quantified this directly: in trials with both placebo and no-treatment arms, much of what is conventionally attributed to the placebo response was indistinguishable from regression to the mean and natural-history drift. The lumped analysis absorbs this regression into the treatment-effect estimate and the analyst has no formal mechanism to recover the true pharmacological effect.

[6]

Biomarker validation is invalidated when a biomarker correlates with PB or TV, rendering precision-medicine stratification wrong regardless of trial size.

- A biomarker correlated with placebo responsiveness or natural-history trajectory is indistinguishable from a true pharmacological predictor in a lumped analysis.
- The PATH statement defers mechanistic attribution to domain knowledge; the BR-PB-TV decomposition supplies that substrate.
- Where such coupling cannot be ruled out, rigorous pharmacological-biomarker validation requires the decomposition; where the biomarker is known orthogonal to PB and TV, the lumped analysis already recovers the pharmacological slope.

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The third failure mode is the biomarker-validation failure. The precision-medicine question that motivates many recent N-of-1 trials is whether a baseline biomarker predicts treatment response, formalised in the PATH statement[14,15]. A biomarker that correlates with placebo responsiveness or with a favourable natural-history trajectory will appear in a one-component analysis as a predictor of the lumped 'treatment effect', indistinguishable from a true pharmacological predictor. The biomarker-stratified prescribing decision that follows from such an analysis is wrong, and the trial's regulatory utility is degraded. The PATH framework explicitly defers the mechanism of treatment-effect heterogeneity to domain knowledge[16,17]; the BR-PB-TV decomposition supplies the mechanistic substrate that PATH leaves blank. We argue that, whenever the candidate biomarker may itself correlate with PB or TV, predictive-biomarker validation cannot be conducted rigorously in a lumped-analysis framework regardless of how large the trial is; the bias is fixed by the biomarker's coupling to the non-pharmacological components, not by sample size. Where the biomarker can be assumed orthogonal to PB and TV, this failure mode does not arise, and the lumped analysis recovers the pharmacological slope, as the simulation of section 7.3 confirms. We stress at the outset the precise form this failure mode takes, because the paper's own simulation history clarifies it. Study A (section 7.3) varies the *magnitude* of PB and TV with the biomarker coupled to BR alone, which the section 6.1 identity predicts to be innocuous, and it finds no bias. The decisive experiment is the contaminated-biomarker pilot (section 7.4), which couples the biomarker to PB and exhibits the failure directly: the lumped bias grows with the biomarker-PB correlation. The operative quantity is the biomarker's correlation with the individual-level PB *variance* ($c_{bm,PB}\,\sigma_{PB}$), not the population-mean placebo response m_{PB}, which does not enter the slope; a trial population can have a large average placebo response and yet pose no validation hazard, while one with zero average but a biomarker-correlated spread of placebo responsiveness is at risk. The failure-mode claims of this section are therefore consequences of the identity, demonstrated in simulation (section 7.4), and the bias is recoverable: Study B (section 7.5) shows that a belief-covariate decomposition recovers the unbiased pharmacological slope under contamination, but only on a design that decouples drug exposure from belief, not on the standard hybrid design.

1.3 The current state of the N-of-1 literature

[7]

The N-of-1 literature has matured substantially but motivates three observations that the present paper addresses.

- The design has evolved from individual-patient consultation to a framework supporting population-level inference.
- CONSORT reporting standards and quality audits document progressive improvement in analytic transparency.
- Three specific gaps in the literature motivate the decomposition framework proposed here.

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The N-of-1 trial in its modern randomised form originates with[18], who defined the within-patient multiple-crossover RCT, demonstrated it on a single airflow-limitation case, and established a clinical service to deliver it at scale. The design inherits its identifying logic from two older traditions: the crossover trial of classical biostatistics, in which within-subject period contrasts estimate treatment effects under explicit carryover assumptions, and the single-case experimental designs of behavioural science, in which phase reversals (withdrawal and reinstatement of treatment) identify effects within one participant. The blinded-discontinuation contrast central to this paper is a direct descendant of both. The methodological step from the individual-patient trial to population-level inference was taken by[19] and[20], who showed that a series of single-patient trials can be combined through hierarchical meta-analysis to estimate both population treatment effects and individual responses, the statistical foundation on which the aggregated design rests. Over the last fifteen years the design has accordingly been reframed from an individual-patient decision tool into an aggregated, meta-analytic framework supporting population-level inference: the programmatic statement of[21], the methodological reviews of[22], the historical-practice audit of[23], and the more recent syntheses of[24],[25],[26], and[27] trace that trajectory. Reporting standards are codified in the CONSORT extension[28] (the headline statement) and[29] (the explanation-and-elaboration companion); the reporting-quality audits of[30] and its successors document slow but progressive improvement in analytic transparency. Three observations about this literature motivate the present paper.

[8]

No published N-of-1 trial employs a structural decomposition of treatment response despite three decades of methodological literature.

- Audit data show the field has relied on paired t-tests, binary-indicator LME, and hierarchical Bayesian regressions.
- Recent methodological advances (Liao 2023, Blackston 2019, Carrozzo 2025) retain the lumped-response framing.
- The decomposition gap persists even as carryover and identifiability problems are increasingly recognised.

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The first observation is that the dominant analytic models are simple. The reporting-quality audit of[30] documents that the published N-of-1 trials between 1985 and 2013 relied predominantly on paired t-tests, simple linear mixed-effects models with a binary treatment indicator, or hierarchical Bayesian regressions. On our own reading of that corpus and the methodological literature that followed it, no published N-of-1 trial has employed a structural decomposition of treatment response into pharmacological, expectancy, and natural-history components. The most recent methodological advances retain this character:[31] develops Bayesian distributed-lag models with autocorrelated errors as a current carryover-aware analysis, but the model still treats the response as a single quantity to be estimated.[32] and[33] compare aggregated N-of-1 trials with parallel and crossover RCT alternatives by simulation and document carryover-induced type-I-error inflation as a key failure mode, but again at the level of the lumped treatment effect. None of these papers decompose the response.

[9]

Placebo-component separation is widely recognised but almost entirely unaddressed at the protocol level in N-of-1 methodology.

- The Hendrickson hybrid blinded-discontinuation phase is among the few protocol mechanisms that supply PB-versus-BR identifiability.
- Placebo run-in, open-label-placebo, and mediator-analysis approaches have not been imported into N-of-1 designs.
- The gap between awareness of the problem and protocol-level solutions motivates the present framework.

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The second observation is that the placebo-component separation problem is widely known but largely unaddressed in N-of-1 specifically. The blinded discontinuation phase used in the Hendrickson hybrid design[34] is one of the few protocol-level mechanisms in routine N-of-1 use that supplies identifiability of PB versus BR. Other relevant approaches, including placebo run-in designs, open-label-placebo trials, and mediator-analysis applications, have been developed in parallel but rarely imported into N-of-1 methodology.

[10]

Kaptchuk's within-placebo decomposition is the closest structural analogue; no published paper applies the corresponding full-response decomposition to N-of-1 trials.

- Kaptchuk et al. (2008) empirically decomposed the placebo response into observation, ritual, and relationship components.
- Open-label placebo trials confirm that clinically meaningful PB-only response is possible even when patients know they receive placebo.
- The BR-PB-TV framework is structurally analogous but operates

at the level of the full treatment response.

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The third observation is that the closest precedent in the literature is Kaptchuk's three-component decomposition of the placebo response itself. In a randomised trial of acupuncture-style placebo for irritable bowel syndrome,[5] decomposed the placebo response into observation, ritual, and patient-practitioner relationship components, demonstrating empirically that the placebo response is itself a sum of identifiable parts rather than a single 'expectancy' lump. The follow-up open-label placebo trials of[35],[36], and[37] in adolescent functional pain establish that even disclosed-placebo intervention can produce clinically meaningful PB-only response. The framework we propose is structurally analogous to Kaptchuk's decomposition but applied at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone. To the best of our knowledge, no published paper proposes the corresponding decomposition for the aggregated N-of-1 setting, even though the methodological ingredients, including mixed-effects modelling, blinded-phase contrasts, and trajectory parameterisation, are all in routine use.

[11]

The decomposition gap is real, the tools to fill it are mature, and the failure modes are large enough to make it a methodological priority.

- The gap between available methodology and current practice is documented and not attributable to missing tools.
- Failure modes in chronic-condition pharmacology are consequential, not academic.
- The remainder of the paper develops, applies, and evaluates the decomposition framework.

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The case we develop in the rest of this paper is therefore that the gap is real, that the methodological tools to fill it are mature, and that the failure modes of the lumped analysis are large enough in chronic-condition pharmacology to make the decomposition a methodological priority rather than an academic refinement.

1.4 The neurobiological and epidemiological reality of the components

[12]

Each component of the decomposition reflects a real causal mechanism documented in independent literatures.

- BR corresponds to pharmacological action; PB to neurobiologically real belief-driven mechanisms; TV to natural-history drift.
- The components are not statistical constructs but distinct causal pathways with separate literatures.
- The section reviews the evidence for each component before establishing the formal model.

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The decomposition is not a statistical convenience. Each component reflects a real causal mechanism documented in independent literatures, and we shall briefly recall each in turn before turning to the formal model.

[13]

BR is the pharmacological target of regulation; PB is neurobiologically real and phase-modulated, which is precisely the design contrast that makes the decomposition identifiable.

- PB is mediated by endogenous opioid/dopamine release, HPA-axis regulation, and top-down attentional shifts.
- Wager et al. neurologic pain signature provides a clean fMRI dissociation between pharmacological and expectancy contributions.
- Phase-dependent belief variation (open-label vs. blinded) is the design contrast that supplies PB identifiability.

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The biological response component represents the pharmacological action of the active compound: the chemical binding, downstream signalling, and physiological response that a regulatory authority cares about when it grants approval. The placebo-belief response reflects neurobiologically real mechanisms documented across pain, depression, Parkinson's, and PTSD research[38–41], mediated by belief-driven release of endogenous opioids and dopamine, anticipatory regulation of the hypothalamic-pituitary-adrenal axis, and top-down attentional shifts in symptom perception. Direct neuroimaging evidence is instructive here:[42] developed an fMRI-based 'neurologic pain signature' (NPS) that tracks nociceptive input and is reduced by the opioid agonist remifentanil, and the individual-participant meta-analysis of[43] found that placebo treatments produce moderate reductions in reported pain but only negligible reductions in the NPS, providing a neural dissociation between the pharmacological and expectancy contributions to a clinical pain response. The placebo response is heterogeneous across patients with biological correlates[9,10] and is modulated by trial phase: when belief varies, as it does between open-label and blinded-discontinuation conditions, the placebo response varies, and the modulation is exactly the design contrast that makes the decomposition identifiable.

[14]

TV is the untreated counterfactual trajectory; its heterogeneity is the central object of precision-medicine inference and must be modelled, not absorbed into noise.

- TV sign depends on disease class: positive for acute recovery, near-zero for stable chronic, negative for progressive conditions.
- Selection-driven enrolment at peak severity guarantees positive TV in early trial weeks regardless of disease class.
- The PATH statement identifies treatment-effect heterogeneity modelling as mandatory for precision-medicine inference.

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The time-variant natural-history component reflects whatever trajectory the participant would have followed without treatment. For acute conditions this trends positive (recovery on intrinsic timescale); for chronic stable conditions it is near zero on the population mean but with substantial participant-level variance; for chronic progressive conditions it trends negative; and for selection-driven enrolment processes it is positive in the early weeks of the trial regardless of underlying disease class, because of regression to the mean[11,13]. The PATH statement[14,15] highlights that this heterogeneity is itself the central object of precision-medicine inference: heterogeneity of treatment effect must be modelled, not absorbed into noise, if the precision-medicine question is to be answered.

1.5 What this paper contributes

[15]

The paper makes four contributions spanning formalisation, identifiability mapping, tractable methodology, and quantification of the inferential cost of lumping.

- Sections 2–3 formalise the decomposition; section 4 maps it to design contrasts; section 5 develops tractable analysis methods.
- Sections 6–7 provide worked examples and a Monte Carlo simulation study; section 8 applies the framework to biomarker validation.

- A companion pedagogical document provides extended worked examples for first-time readers.

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We make four contributions, each addressed in a section below. We begin in section 2 with the formal model and notation. Section 3 develops each component in turn, with biological motivation, parametric form, and identifiability conditions. Section 4 maps the design-contrast structure of the standard hybrid open-label-plus-blinded-discontinuation design onto the identifiability requirements of the decomposition. Section 5 develops the analysis-side methodology, including the linear-mixed-effects approximations and the phase-by-treatment indicator extension that retains BR-PB sensitivity at low parametric cost. Section 6 presents the worked examples, and section 7 presents the simulation study. Section 8 discusses applications to predictive-biomarker validation. The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` provides an extended worked-example treatment for readers approaching the material for the first time.

2 Notation and the formal model

[16]

This section fixes compact notation and states the formal model; a fuller worked-example treatment is in the companion pedagogical document.

- Definitions are precise enough to support the identifiability and analysis-side arguments in later sections.
- The presentation is kept compact; extended derivations are deferred to the companion document.
- The companion pedagogical document provides numerical worked examples at each formal step.

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In this section we shall fix notation and state the formal model. The three components are defined precisely enough to make the identifiability and analysis-side arguments in later sections rigorous, but we have endeavored to keep the presentation compact. A fuller treatment, with worked numerical examples at each step, is available in the companion pedagogical document at `docs/24-component-decomposition-pedagogy.md`.

[17] where Y_{it} is the observed symptom score, BL_i is participant i's

baseline score, the three components BR, PB, TV are non-negative reductions in symptom severity attributable to the three causal mechanisms identified in section 1, and ε_{it} is observation-level noise. The sign convention is that an increase in any component reduces Y; all three components are zero at $t = 0$ by construction (no exposure yet to drug, no belief activation, no elapsed natural history) and grow over the duration of the trial.

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For participant i at trial timepoint t:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[\, BR_{it} + PB_{it} + TV_{it} \,\right] \;+\; \varepsilon_{it},$$

[18]

Each component is driven by its own time-on-state variable, making the components structurally distinct even when they overlap in calendar time.

- BR_{it} accumulates with time-on-drug; PB_{it} with time-on-belief scaled by expectancy $\eta(\text{phase})$; TV_{it} with calendar time from trial entry.
- The distinct time indices are the structural source of identifiability from phase-contrast data.
- All components are zero at $t = 0$ by construction and grow over the trial duration.

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Each component is a participant-specific function of its own time-on-state variable. BR_{it} depends on time-on-drug, the cumulative exposure to active medication since drug initiation. PB_{it} depends on time-on-belief, modulated by a phase-specific expectancy factor $\eta(\text{phase})$ that captures the patient's confidence that they are receiving active treatment. TV_{it} depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

[19] rescaled so that $C(0) = 0$ and $C(t) \to m_C(1 - e^{-d_C})$ as $t \to \infty$, for $C \in \{BR, PB, TV\}$. The three Gompertz parameters (m_C, d_C, r_C) are participant-specific random effects drawn from population distributions, and the participant-level heterogeneity in these distributions is the principal source of between-person response variability that the framework is designed to characterise.

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Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

Term	Meaning
BR	Biological response: pharmacological reduction in symptom score attributable to active drug.
PB	Placebo-belief response: reduction attributable to the patient's belief that they are receiving treat
TV	Time-variant component: reduction (or increase) attributable to the natural-history trajectory of
η(phase)	Expectancy multiplier for PB: 1 in open-label on-drug, ≈ 0.5 in blinded discontinuation, between
BL_i	Participant i's baseline (pre-treatment) symptom score.
D_{it}	Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug.
$t_{sd,it}$	Time since drug discontinuation (off-drug only).
λ	Carryover decay rate, with half-life $t_{1/2} = \ln 2/\lambda$.

Each component follows a modified Gompertz curve:

$$C(t) \;=\; m_C\,\exp(-d_C\exp(-r_C t)) \;-\; m_C\,\exp(-d_C),$$

[20]

The residual term has AR(1) structure and the three components are permitted to co-vary through cross-component covariances.

- Within-participant residuals follow $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$.
- Within-time and across-time cross-component covariances permit participant-level heterogeneity in one component to co-vary with the others.
- This structure accommodates biologically plausible correlations, such as high-BR patients also tending toward high-PB.

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ε_{it} is a within-participant residual term with AR(1) serial-correlation structure $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$. The three components are permitted to be jointly correlated through within-time and across-time cross-component covariances c_{cf1t} and c_{cfct} respectively, so that participant-level heterogeneity in any one component can co-vary with heterogeneity in the others.

The principal psychometric and pharmacological terms used below are summarised in Table 1.

3 The three components

[21]

This section presents each component with its biological motivation, Gompertz parameterisation, and identifiability conditions to establish the model as causally interpretable.

- The three components are presented in turn: BR (biological response), PB (placebo-belief), TV (natural history).
- Each component is grounded in independent biological literature before its parametric form is stated.
- The section's purpose is to demonstrate that the model sum reflects causal structure, not mathematical convenience.

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In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterisation, and the design contrasts that identify it. The treatment is organised so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

3.1 Biological response (BR)

[22]

BR is the pharmacological target of regulatory approval and biomarker stratification, independent of belief or natural history.

- BR arises from chemical action on the body and is zero for inert pills regardless of patient expectation.
- It is the estimand that approval decisions, dose selection, and biomarker-guided stratification require.
- Separating BR from PB and TV is the central precision-medicine problem the framework addresses.

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BR is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief, suggestion, and natural history: a participant who somehow received active drug without knowing it would still develop a measurable BR, and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, BR is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around BR.

[23]

PK/PD considerations produce a saturating BR trajectory with a slow initial rise, a peak-growth phase, and a sharp drop to zero on discontinuation.

- Plasma steady state is reached within a few half-lives; pharmacodynamic integration (e.g. receptor adaptation) adds additional lag.
- For prazosin-PTSD, Raskind et al. (2013) calibrate the onset to several weeks of sustained exposure.
- The saturating shape motivates the Gompertz over linear or simple exponential alternatives.

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The qualitative shape of the BR trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life; brain concentration follows plasma with a short delay; the downstream pharmacodynamic response (reduction in symptom-relevant neural activity, in the case of prazosin's effect on noradrenergic tone during REM sleep) typically requires several days to weeks of sustained exposure to integrate fully[44]. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life $t_{1/2} = \ln 2/\lambda$ rather than an instantaneous drop. The rate of this off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

[24]

The three Gompertz parameters (m_{BR}, r_{BR}, d_{BR}) capture ceiling, onset rate, and climb shape respectively, calibrated to prazosin-PTSD reference values.

- m_{BR} is the biological ceiling (maximum reduction at infinite exposure); r_{BR} the onset rate; d_{BR} the displacement shaping the early climb.
- Prazosin-PTSD calibration yields $m_{BR} \approx 11$, $r_{BR} \approx 0.42$/week, $d_{BR} \approx 5$.
- These are participant-specific random effects whose population distributions encode between-person response heterogeneity.

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The modified Gompertz form $BR(t) = m_{BR} \exp(-d_{BR} \exp(-r_{BR} t))-$

$m_{BR}\exp(-d_{BR})$ captures this profile compactly. The asymptote $m_{BR}(1-e^{-d_{BR}})$ is the biological ceiling, the largest reduction the drug can produce at infinite exposure; the scale parameter m_{BR} equals that ceiling up to the factor $(1-e^{-d_{BR}})$, which for the calibrated $d_{BR} \approx 5$ exceeds 0.99. The onset rate r_{BR} controls how rapidly the participant approaches the ceiling: high r_{BR} means a fast onset (effect plateaus in one to two weeks), low r_{BR} a slow climb (six to eight weeks). The displacement d_{BR} shapes the climb: high d_{BR} gives an early-onset curve that rises sharply at first, low d_{BR} a more gradual climb. For PTSD prazosin, calibrated values from the[34] reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at roughly four to five weeks of sustained exposure), and $d_{BR} \approx 5$ (moderately steep early climb).

[25]

The Gompertz is preferred over linear and simple-exponential alternatives because it captures the asymmetric slow-start saturation profile of chronic-condition pharmacodynamics.

- Linear $BR = \beta t$ is biologically implausible past the first few weeks due to unbounded growth.
- Exponential $BR = m(1-e^{-rt})$ lacks the slow-start phase from receptor adaptation and gene-expression changes.
- The Gompertz interpolates between both limits and its three-parameter form avoids overfitting; sensitivity analysis is reported separately in `07-gompertz-evaluation`.

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Two alternative families that do not capture the asymmetric saturation profile are worth noting for contrast. A linear function $BR = \beta t$ climbs without bound and is biologically implausible past a few weeks. An exponential decay $BR = m_{BR}(1-\exp(-rt))$ is acceptable for fast-onset drugs but lacks the slow-start phase that characterises most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high d limit) and linear-in-t growth (low d limit), and its three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting. The Gompertz is a deliberate choice over the two parametric forms most familiar from pharmacokinetic-pharmacodynamic modelling. The sigmoid $E_{\max}$ (Hill) model[45] describes the dependence of effect on concentration (a dose-response curve), whereas $BR(t)$ here is a time-course of accumulating effect at fixed dosing; and the indirect-response (turnover) models that do describe effect time-courses are specified through differential equations on a hidden response variable rather than in closed form. The modified Gompertz retains the slow-start, saturating shape those models produce while remaining a closed-form, three-parameter mean function that is tractable inside a mixed-effects likelihood. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is reported separately as `07-gompertz-evaluation`. The headline result of that evaluation is reassuring for the present framework: in a 1{,}000-replicate-per-cell simulation that generated data under the logistic, hyperbolic-tangent, and piecewise-linear breakpoint families and analysed them under the standard linear-mixed model, the trajectory family was essentially irrelevant to the power of the biomarker-treatment interaction test (the family-to-family spread fell within two Monte Carlo standard errors of sampling noise) and to type I error (0.025 to 0.036 across null cells, with no family-specific inflation). The conclusions drawn here are therefore not artefacts of the Gompertz form, though the regimes the family cannot represent (cyclical or biphasic trajectories) remain genuine exclusions.

[26]

BR is identified by the blinded-discontinuation contrast, where drug removal collapses BR to zero while PB remains intact.

- Within-participant on-drug versus off-drug contrasts are the primary identification mechanism.
- The blinded-discontinuation phase provides the cleanest contrast: BR drops, PB is preserved.
- This contrast does not require the analyst to model PB or TV explicitly to recover BR.

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The BR component is identified by within-participant on-drug versus off-drug contrasts, formalised in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, BR decays toward zero as the drug clears while PB remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of BR that does not require the analyst to model PB or TV explicitly. The decay is not instantaneous: with the carryover half-life $t_{1/2} = \ln 2/\lambda$ of the analysis-side decay term, residual BR persists into the early off-drug observations, so the contrast is clean only once the discontinuation window is long relative to $t_{1/2}$, and the drug state is best carried by the continuous-decay indicator D_{it} of Table 1 rather than a hard on/off indicator.

3.2 Placebo-belief response (PB)

[27]

PB is the strict placebo response: improvement that persists when the active drug is silently replaced, provided the patient's belief is unchanged.

- PB is belief-dependent, not drug-dependent; it would survive a silent switch to placebo under sustained belief.
- It is distinct from BR in causal origin: BR requires chemical action; PB requires only belief.
- The definition is strict and operational, not vague or colloquial.

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PB is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

[28]

PB reflects neurobiologically real mechanisms triggered by belief, not pharmacology; neuroimaging and genetic evidence confirm it is a structured biological phenomenon.

- Belief-driven endogenous opioid/dopamine release, HPA-axis modulation, and top-down attentional shifts are all documented mechanisms.
- Wager et al. neurologic pain signature dissociates pharmacological from expectancy contributions via fMRI.
- COMT genotype predicts placebo responsiveness, confirming biological structure rather than measurement artefact.

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It is worth being precise about what PB is and is not. PB is not 'fake' improvement; it is a phenomenon clinicians knowingly exploit, a national survey of US internists and rheumatologists finding roughly half report regular use of placebo treatments[46]. As the introduction sets out, the placebo response is mediated by well-characterised neurobiological pathways[38–41,47]: belief-driven release of endogenous opioids and dopamine, anticipatory hypothalamic-pituitary-adrenal regulation, and top-down attentional shifts, dissociated from pharmacology by the[42] neurologic pain signature[43] and shown to have genetic correlates[9,10]. What makes these mechanisms 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology.

[29]

PB is identifiable from BR only when belief and treatment allocation differ, as in the blinded-discontinuation phase.

- The blinded-discontinuation phase is the canonical contrast: patients cannot tell whether they remain on active drug.
- Belief drops to a partial level ($\eta \approx 0.5$) because the patient's expectation is hedged, not absent.
- Designs without a phase that decouples belief from treatment cannot separately identify PB from BR.

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For trial design, the relevant point is that PB is separately identifiable from BR if and only if the trial includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in treatment correspondingly drops to a partial level (multiplier roughly 0.5) because the patient's expectation is hedged.

[30]

The $\eta = 0.5$ blinded-phase expectancy is a modelling assumption operationalising response expectancy, not a measured quantity, and requires sensitivity analysis.

- The construct is response expectancy in the sense of Kirsch (1985); to our knowledge the residual expectancy under blinded discontinuation has not been measured directly.
- Open-label placebo trials confirm clinically meaningful PB-only responses even under explicit disclosure, bounding the residual response away from zero.
- Sensitivity analyses varying η are mandatory before clinical conclusions depend on this single parameter.

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PB also follows a Gompertz curve, scaled by the expectancy factor:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB}\, \exp(-d_{PB} \exp(-r_{PB} t)) + (\text{offset to start at zero}).$$

Typical expectancy values are $\eta = 1.0$ in open-label on-drug, $\eta \approx 0.5$ in blinded discontinuation, and η between 0 and 0.5 in open-label placebo (the residual expectation a patient retains when told the previous active phase was beneficial). The quantity η operationalises response expectancy in the sense of[48], the patient's anticipation of benefit that the expectancy literature treats as a determinant of experienced symptom change. The choice of $\eta = 0.5$ for the blinded phase is a modelling assumption rather than a calibrated quantity: to our knowledge the residual expectancy a patient carries when uncertain about allocation has not been measured directly as a fraction of the open-label level, and we adopt 0.5 as a deliberately central placeholder. The available indirect evidence is nonetheless that the residual response is substantial. Open-label placebo trials[35–37] demonstrate that a clinically meaningful response persists even when patients are explicitly told they are receiving an inert intervention, which bounds the residual belief-driven response away from zero. Because the value of η is not empirically pinned, sensitivity analyses varying it across its plausible range are mandatory before any conclusion is allowed to depend on this single number.

[31]

Kaptchuk et al. (2008) provide the structural precedent; the BR-PB-TV framework extends the same decomposition logic to the full treatment response.

- Kaptchuk decomposed the placebo response itself into observation, ritual, and relationship components empirically.
- The present framework applies analogous decomposition logic at the level of drug, placebo, and natural history.
- Both frameworks reject the 'expectancy lump' model in favour of identifiable additive parts.

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The decomposition framework as proposed here has a structural analogue in the placebo-research literature.[5] decomposed the placebo response itself into observation, ritual, and patient-practitioner relationship components in a randomised trial of acupuncture-style placebo for irritable bowel syndrome, demonstrating empirically that the placebo response is not a single 'expectancy' lump but a sum of identifiable parts. The present BR-PB-TV framework is analogous in spirit but operates at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone.

[32]

PB variance scales with η, not just its mean; ignoring this heteroscedastic structure miscalibrates standard errors and downstream biomarker tests.

- Phase-dependent variance follows $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$.
- Open-label phases have high between-patient PB variance; blinded phases have low variance.
- Ignoring heteroscedasticity deflates standard errors in open-label and inflates them in blinded phases, with consequences for CI coverage and biomarker test calibration.

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A subtle property of PB is that its variance also scales with η, not just its mean. When η is high, there is substantial variability across patients in how strongly the placebo response is expressed; high-suggestibility patients show large PB, low-suggestibility patients show essentially none. When η is low (blinded), the placebo response is uniformly attenuated and the patient-to-patient variance shrinks accordingly. The model therefore parameterises $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic structure deflates standard errors in the open-label phase and inflates them in the blinded phase, with downstream consequences for confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

3.2.1 Additivity, subadditivity, and the BR-PB interaction

[33]

Additivity is a modelling assumption, not biological fact; the section examines whether a $BR \times PB$ interaction term is warranted.

- The formal model writes total response as a sum; this section asks whether the BR-PB interaction is zero.
- The question has inferential consequences for biomarker validation and for what the phase-augmented analysis captures.
- The treatment is preparatory for the generalised interaction model introduced later in this section.

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In this section we shall broach a question that the formal model leaves open: whether the three components contribute additively to symptom reduction, or whether there is a meaningful interaction between BR and PB that the additive specification misses.

[34]

The additive model may be wrong: published literature documents subadditivity between pharmacological and placebo-belief effects in some clinical contexts.

- Subadditivity means the combined $BR + PB$ effect is less than the sum of each component measured separately.
- Evidence from Kaptchuk et al. and Benedetti et al. shows the phenomenon is context-dependent, not universal.
- Formal test: $\partial Y/\partial(BR \cdot PB) > 0$ in a saturated mean-structure model.

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The decomposition as written in section 2 treats the three components as additive contributors to symptom reduction:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; (BR_{it} + PB_{it} + TV_{it}) \;+\; \varepsilon_{it}.$$

The additivity is a modelling assumption, not a fact about biology, and it is worth stating that the assumption may be wrong in directions the framework can absorb and in directions it cannot. Additivity is also scale-dependent in the sense emphasised by[49]: components that combine non-additively on the raw symptom scale may combine additively on a transformed (for example logarithmic) scale, so an apparent $BR \times PB$ interaction can be partly an artefact of the chosen response metric rather than a biological fact. That belief and pharmacology interact rather than act in isolation is demonstrated directly by[50], who showed with neuroimaging that positive treatment expectation roughly doubled the analgesic benefit of the opioid remifentanil while negative expectation abolished it, so the two channels are not in general separable by simple addition. More specifically, the published literature on placebo-drug interactions has accumulated evidence that the combined effect of receiving an active drug and believing one is receiving an active drug is, in some clinical contexts, less than the sum of the two components measured separately[5,38]. The phenomenon is typically termed subadditivity: $\partial Y/\partial(BR{\cdot}PB) > 0$ when both components are present and their interaction term is included in a saturated mean-structure model.

[35]

Three candidate mechanisms for subadditivity are shared neural substrate, scale ceiling effects, and belief-dynamics adaptation.

- Shared pain-modulation circuits produce saturating occupancy when both BR and PB activate them simultaneously.
- Scale ceiling effects prevent registration of additional symptom reduction once $BR + PB$ saturates the measurement range.
- Belief-dynamics adaptation: unexpectedly large BR can recalibrate patient expectation downward, attenuating ongoing PB.

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Three mechanisms have been proposed in the placebo-research literature to explain subadditivity. First, a shared neural substrate: if the descending pain-modulation circuits that mediate placebo analgesia overlap with the circuits modulated by the active drug, simultaneous activation by both mechanisms produces less marginal benefit than activation by either alone, in a saturating-occupancy sense. Second, ceiling effects on the symptom scale: a clinical scale that caps at zero (no symptoms) cannot register additional reduction once a participant has reached that cap, so any participant whose $BR + PB$ already saturates the scale will register no further reduction even if BR or PB would, in isolation, predict more. Third, belief-dynamics adaptation: the patient's expectation may adapt downward when the active drug produces a strong response that exceeds the expected effect, reducing the ongoing PB contribution; equivalently, unexpectedly large BR may attenuate PB via expectation recalibration.

[36] where $\gamma < 0$ encodes subadditivity, $\gamma = 0$ recovers the additive form, and $\gamma > 0$ encodes superadditivity, is the natural extension. The interaction coefficient γ is identifiable from the same trial-design contrasts that identify the marginal components: open-label phases supply $BR \cdot PB$ products at high η, blinded phases supply $BR \cdot PB$ products at attenuated η, and the γ coefficient is estimated from the difference between the two regimes once the marginal components are accounted for.

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Whichever mechanism dominates in a given clinical context, the subadditivity question is empirical and design-dependent. Within the BR-PB-TV framework, subadditivity manifests as a non-zero coefficient on a $BR \times PB$ interaction term in the conditional-mean specification. The additive form treats this interaction as zero by construction; a generalised form

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; [BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}] \;+\; \varepsilon_{it},$$

[37]

Subadditivity biases biomarker predictive accuracy estimates and is partly absorbed by the phase-augmented analysis even without explicit γ estimation.

- Under subadditivity, the biomarker-by-BR association is partly cancelled, inflating apparent predictive accuracy in the additive model.
- Phase designs attenuating PB (blinded, crossover) push the effective biomarker-treatment estimate closer to the true biomarker-by-BR value.
- The phase-augmented analysis in section 5 absorbs subadditivity as a side benefit.

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For trial design, two consequences follow. First, a biomarker that predicts BR in an additive-decomposition analysis may overstate or understate predictive accuracy if the true mechanism is subadditive: under subadditivity, the biomarker-driven increase in BR is partly cancelled by the $\gamma \cdot BR \cdot PB$ interaction, and the marginal biomarker-treatment association is smaller than the biomarker-by-BR association. Second, trial designs that attenuate PB in some phases (blinded discontinuation, crossover) implicitly down-weight the subadditivity contribution to the response in those phases, which means that the effective biomarker-treatment association estimated from those phases is closer to the biomarker-by-BR association than is the open-label estimate. The phase-augmented analysis recommended in section 5 below therefore has the side benefit of partly absorbing subadditivity, even when γ is not estimated explicitly.

[38]

A forthcoming simulation study will quantify how bias in the biomarker-treatment interaction scales with γ; the pre-registration predicts linear scaling.

- Bias in $\hat{\beta}_{bm:D}$ is expected to scale linearly with γ at small $|\gamma|$.
- Under subadditivity, bias sign is opposite to the true biomarker-by-BR effect; under superadditivity it is aligned.
- Empirical quantification is deferred to the production simulation programme in section 7.

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A simulation study quantifying the magnitude of bias in the biomarker-treatment interaction estimate when the additive analysis is fitted to data generated under non-zero γ is forthcoming as part of the simulation programme described in section 7. The pre-registration is that bias scales linearly with γ at small $|\gamma|$, with the sign opposite to the true biomarker-by-BR effect under subadditivity and aligned with it under superadditivity.

[39]

Non-zero $\hat{\gamma}$ does not establish true subadditive interaction; latent-class structure with class-correlated components produces the same marginal pattern.

- If class membership correlates with both BR and PB strength, the marginal covariance of $BR \cdot PB$ mimics a direct interaction term.
- This confounding is invisible from the lumped-analysis perspective but is identifiable in a class-aware analysis.
- Apparent subadditivity can arise from latent population heterogeneity with no direct BR-PB interaction in the class-conditional DGP.

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A subtle but inferentially important point merits attention here: a non-zero $\hat{\gamma}$ recovered from an additive-extension analysis does not, on its own, establish that the underlying mechanism is subadditive interaction in the strict sense. The same marginal pattern can arise from latent population structure that the analyst is not modelling. If the trial population is a mixture of treatment-responders and non-responders (in the latent-class sense developed at length in `analysis/report/03-latent-class-mixture-application/`), and class membership correlates with both BR strength and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function $\pi(B)$. From the lumped-analysis perspective this looks exactly like a γ-weighted $BR \times PB$ interaction even though no such direct interaction was specified in the class-conditional DGP. That is, the latent-class structure is one mechanism that produces apparent subadditivity in a one-component analysis, regardless of whether the underlying biology has a true direct BR-PB interaction.

[40]

Distinguishing true subadditivity from latent-class confounding requires either a class-aware analysis or the phase contrasts the BR-PB-TV framework provides.

- The two mechanisms produce qualitatively similar marginal effects but are inferentially distinct.
- $\hat{\gamma}$ should be reported alongside a class-aware sensitivity analysis, not as standalone evidence for mechanistic subadditivity.
- The latent-class paper's `pb_class_correlation` simulation axis is the direct test; cross-paper synthesis is the cleanest disentanglement.

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The two mechanisms, namely true subadditive interaction and latent-class structure with class-correlated components, produce qualitatively similar marginal effects but are inferentially distinct. Distinguishing them requires either a class-aware analysis that explicitly models the latent population structure (as the latent-class paper develops at length), or a phase contrast that separates the mechanisms by exploiting the differential expectancy modulation of PB that the BR-PB-TV framework provides. Under the additive-extension analysis fitted to mixture-DGP data with class-correlated components, $\hat{\gamma}$ is identified up to confounding with the class structure, and the analyst should report $\hat{\gamma}$ alongside a class-aware sensitivity analysis rather than as standalone evidence for mechanistic subadditivity. The Study 1 simulation in the latent-class paper includes a `pb_class_correlation` axis that varies $\mathrm{Cor}(BR, PB)$ across class labels; that axis is the direct simulation-side test of the latent-class generation of apparent subadditivity, and the cross-paper synthesis between the two papers is the cleanest way to disentangle the two mechanisms.

3.3 Time-variant natural-history component (TV)

[41]

TV is the untreated counterfactual trajectory; it is participant-specific, structured, and can be positive, negative, or near-zero depending on disease class.

- TV is the symptom change that would have occurred without treatment, integrated over the trial duration.
- It varies in sign across disease classes and across patients within a class.
- It is independent of treatment status and belief, distinguishing it structurally from PB.

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TV is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

[42]

TV is a structured participant-specific signal, not noise; its distinguishing feature from PB is independence from treatment status and belief.

- Unlike ε_{it}, TV is participant-level (slow trajectory) and requires explicit modelling, not absorption into noise.
- Off-drug baselines and follow-up windows are the primary data sources for estimating TV.
- TV is modulated by neither treatment assignment nor patient belief, whereas PB is modulated by both.

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Like PB, TV is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

3.3.1 When does natural history actually exist?

[43]

The shape of TV depends on disease class, trial timescale, and selection process; none of these can be assumed and all must be recovered empirically.

- The assumption that subjects improve over time is true for some conditions and systematically wrong for others.
- Disease class, trial timescale, and enrolment selection jointly determine the sign and magnitude of TV.
- Recovering the empirical shape of TV is one of the decomposition's primary objectives.

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A common simplifying assumption in trial analysis is that subjects will improve over time, with or without treatment. This is true for some conditions and problematic for others. The shape of TV depends on the underlying disease class, the trial timescale, and the selection process by which participants entered the trial. None of these can be assumed without thought, and the empirical shape of TV is one of the things the decomposition is designed to recover.

[44]

TV sign varies systematically by disease class: positive for acute recovery, near-zero for stable chronic, negative for progressive, and structured for cyclical conditions.

- Acute conditions (post-operative pain, acute back pain) resolve intrinsically; substantial TV-improvement appears in the placebo arm.
- Progressive conditions (Alzheimer's, ALS) require explicit negative-trajectory modelling; apparent treatment effect is slowing of progression.
- Cyclical/triggered conditions (PTSD with trauma anniversaries, relapsing-remitting MS) require covariates beyond the Gompertz form.

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Acute conditions tend toward TV-positive: a sprained ankle, a post-operative pain syndrome, an episode of acute back pain all resolve on a timescale of days to weeks even without intervention, and a trial of an analgesic in such a setting will see substantial TV-improvement in the placebo arm. Chronic stable conditions (treated hypertension, controlled chronic asthma, long-standing PTSD without active triggers) tend toward TV-zero on the population mean but with substantial individual variance. Chronic progressive conditions (Alzheimer's disease, ALS, late-stage congestive heart failure, many cancers) tend toward TV-negative: a trial of a neuroprotective agent in early Alzheimer's must explicitly model the negative natural-history trajectory, since the apparent treatment effect is the slowing of progression, not absolute improvement. Cyclical and triggered conditions (PTSD with trauma anniversaries, seasonal-pattern major depression, relapsing-remitting multiple sclerosis) show structured TV that is neither monotone nor stationary, and may require explicit covariates beyond the Gompertz form.

[45]

Selection-induced TV is real and rarely zero even for stable chronic conditions, because enrolment coincides with peak severity triggering help-seeking.

- Regression to the mean alone contributes several points of apparent improvement in early trial weeks.
- Hrobjartsson et al. systematic reviews across 200+ trials show much of conventional placebo response is indistinguishable from natural-history drift.
- Kirsch (2008) and Locher (2015) demonstrate the clinical consequences in antidepressant and late-life depression trials.

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Selection-induced TV is real and rarely zero, even when the underlying condition is stable. Patients enrol at moments of clinical engagement that often coincide with recent worsening, so regression to the mean alone produces several points of apparent TV-improvement in the early weeks of a trial. As the introduction documents, the systematic-review evidence of[11],[12], and[13] (the last covering 202 trials across 60 conditions) and the antidepressant analyses of[6] and[7] show that much of what is conventionally attributed to placebo response is indistinguishable from regression to the mean and natural-history drift.

[46]

Trial participation itself induces TV-improvement through Hawthorne, structured-visit, and diary-completion effects, independent of pharmacology or placebo.

- Hawthorne, structured-visit, and diary-completion effects each contribute symptom improvements attributable to participation, not treatment.
- These contributions are absorbed into TV in the decomposition, not into BR or PB.
- Ignoring these participation-induced effects inflates the apparent pharmacological and placebo components.

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Trial participation itself also induces TV: the Hawthorne effect, the structured-visit effect, and the diary-completion effect appear in the data as TV-improvement that is independent of any pharmacological or placebo response.

3.3.2 Can we assume no TV effect?

[47]

Assuming $TV = 0$ is rarely safe in chronic psychiatric or pain trials; the conservative default is to estimate TV from the data.

- Zero-TV assumption is valid only when all four conditions hold: stable disease, short trial, non-severity-peaked recruitment, no ancillary care.
- For most chronic conditions at least one condition fails; the assumption should be tested, not asserted.
- Statistical efficiency lost by estimating TV is smaller than the bias introduced by assuming it zero.

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But how safe is it to assume TV is zero? The short answer is: rarely safe, and never safe without empirical support. The longer answer is that TV may be assumed zero only if the disease is in a stable chronic phase, the trial is short relative to the disease timescale, the recruitment process does not favour participants at peak severity, and the trial protocol does not introduce ancillary care that itself improves symptoms. For most chronic trials in psychiatric or pain conditions, at least one of these conditions fails. The conservative default is to estimate TV from the data, even at the cost of some statistical efficiency, rather than assume it is zero.

[48]

The decomposition makes the zero-TV assumption testable and shows that lumping TV into residuals produces autocorrelated, heteroscedastic residuals that bias BR and PB standard errors.

- $\hat{m}_{TV}$ with a variance estimate from the model provides an empirical test of the zero-TV assumption.
- Participant-level TV is structured and correlated with the on/off-drug schedule; it is not independent noise.
- Lumping TV into ε produces autocorrelated within-participant residuals and heteroscedastic cross-participant residuals.

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The decomposition makes this assumption testable. A model that includes a TV component will return $\hat{m}_{TV}$ with a variance estimate; if both are small relative to $\hat{m}_{BR}$ and $\hat{m}_{PB}$, the analyst has empirical support for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others. Setting TV to zero and lumping it into ε is worse than estimating it: the participant-level TV

is structured (each participant has their own slow trajectory) and not independent of the on-drug-versus-off-drug schedule, so lumping it into noise produces residuals that are autocorrelated within participants and heteroscedastic across them. The resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

3.3.3 Why a Gompertz, not a linear time covariate?

[49]

A linear `week` covariate is insufficient for TV because it enforces linearity, ignores participant heterogeneity, and cannot distinguish TV from BR without the explicit decomposition.

- A `week` covariate constrains each additional week to contribute a fixed natural-history change, misrepresenting curved or saturating disease trajectories.
- A single coefficient estimates a population-mean trajectory, ignoring participant-level heterogeneity in natural history.
- Without the decomposition and design contrasts, `week` absorbs both BR and TV, making them inseparable.

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A natural objection is that a model with `week` as a covariate already controls for time. But how adequate is such a control? Three reasons suggest it is not enough. First, a `week` covariate enforces a linear trajectory, constraining each additional week to contribute a fixed amount of natural-history change; real disease trajectories curve, plateau, or accelerate, and the Gompertz form captures the S-shaped or saturating patterns linear time cannot. Second, a single `week` coefficient estimates a population-mean trajectory that ignores the participant-level heterogeneity in natural history. Third, BR and TV both evolve smoothly over the same trial timescale, and a `week` covariate cannot tell them apart from observational data alone; the design contrasts discussed in section 4 are what supply identifiability, and without the explicit decomposition the `week` coefficient absorbs both.

4 Identifiability and trial-design contrasts

[50]

Identifiability is supplied by trial design; each phase of the hybrid design contributes a different combination of compo-

nents to the observed trajectory.

- The decomposition is mathematically specified but requires trial-design contrasts to be statistically identified.
- This section characterises the hybrid open-label-plus-blinded-discontinuation design as the primary identifiability vehicle.
- Each of the three phases supplies a distinct mixture of BR, PB, and TV.

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The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterise the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

[51]

Three phase-by-allocation contrasts isolate PB, BR, and TV respectively; the LME model combines them but the inference is driven by design.

- Open-label versus blinded contrast yields approximately $0.5 \cdot PB$, isolating the belief component conditional on a known blinded-phase η.
- Blinded on-drug versus blinded placebo yields BR once drug carryover has elapsed (since PB and TV match across arms within the blinded phase).
- Off-drug phases yield TV plus residual off-drug PB, with no active BR contribution beyond carryover.

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Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields

$PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$, which isolates the belief component, but only conditional on a known blinded-phase expectancy η: from a two-phase design just the product $\eta \cdot m_{PB}$ is identified, and separating η from m_{PB} requires the open-label placebo phase. The blinded on-drug versus blinded placebo contrast yields BR once drug carryover has elapsed, since PB and TV match across allocation arms within the blinded phase. All phases off drug yield TV plus any residual off-drug PB, with no active BR contribution beyond carryover and PB at the off-drug expectation level. The $\mathbf{1}_{\text{on drug}}$ indicator in the phase table above is shorthand for the continuous-decay drug state D_{it} defined in Table 1.

This conditioning on η is the framework's principal identification limitation and deserves emphasis rather than a parenthetical. With a two-phase (open-label and blinded) design the data identify only the product $\eta \cdot m_{PB}$, so PB is recovered up to the assumed blinded-phase expectancy; the working value $\eta \approx 0.5$ is a modelling assumption, not an estimate, and absent a belief or expectancy measurement it is not falsifiable from the outcome data alone. The identified set for m_{PB} is accordingly the interval $\{\widehat{\eta m_{PB}}/\eta : \eta \in (0, 1]\}$, which collapses to a point only when η is pinned down. Two design features close the gap: an open-label placebo phase, which supplies a second equation separating η from m_{PB}, and direct measurement of belief, as in the open-hidden and balanced-placebo designs[51,52] that manipulate or record the patient's treatment expectation. We therefore recommend that any trial relying on the decomposition for a PB-versus-BR attribution either include such a phase or collect an expectancy measure, and that analyses report sensitivity of the attribution to η across its plausible range.

[52]

Designs lacking specific phases lose identifiability of the corresponding component; the decomposition machinery is only as useful as the design allows.

- A pure open-label design cannot separate BR from PB because both are present at full strength throughout.
- A pure blinded design cannot estimate $\eta = 1$ PB because no phase achieves full-strength belief.
- Trial designers should verify each component has at least one phase in which it is uniquely present before committing to the design.

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A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

[53]

Design-contrast identifiability is necessary but not sufficient: the three Gompertz trajectories are collinear, so joint estimation is credible only under specific design and sample-size conditions.

- The components share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present.
- Joint recovery requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough for TV to separate from a decaying BR.
- The recoverable region of the design-and-sample-size space is mapped by the pre-registered identifiability diagnostics, not asserted.

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Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for BR, the open-label-versus-blinded contrast for PB, and an off-drug window for TV), and that off-drug window must be long enough relative to the carryover half-life for TV to separate from a still-decaying BR. Where these conditions are weak the components are jointly identified only up to a near-collinear direction, and the symptom of that is the rank-deficiency the pilot of section 7 encountered when the full phase-augmented formula was fitted at $N = 35$. We therefore do not assert that the three components are always recoverable. The pre-registered identifiability diagnostics of section 7 (convergence rates of the component-matched fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the BR estimate to the assumed η(phase) multipliers) are the means by which the recoverable region of the design-and-sample-size space is to be mapped, and the framework's participant-level claims should be read as conditional on falling inside that region.

5 Analysis-side methodology

[54]

Four constraints make a component-matched analysis impractical at trial-relevant sample sizes; a phase-augmented LME captures the important part of the gap at minimal cost.

- The question of whether to mirror the DGP in the analysis model is reasonable; the section addresses it directly.
- The four constraints are identifiability, degrees of freedom, convergence fragility, and marginal estimand robustness.
- A phase-by-treatment indicator is the recommended modest extension that captures BR-versus-PB sensitivity without the full parametric cost.

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In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data? The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

5.1 Four reasons not to match the DGP

[55]

The first constraint is identifiability: forcing a three-component model on a design with insufficient contrasts produces unidentified parameters, not component estimates.

- A pure open-label trial cannot distinguish BR from PB; both rise from zero together on treatment initiation.
- Fitting three components to a one-contrast design produces implausibly tight standard errors that misrepresent inferential precision.

- The analyst must audit whether the design supplies the necessary contrasts before specifying a component-matched model.

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The first reason is identifiability. The trial design must supply the contrasts that identify each component, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV). A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

[56]

The second constraint is degrees of freedom: a component-matched analysis adds approximately a dozen non-linear parameters, inflating moderation estimand variance at $N \in [30, 150]$.

- Gompertz BR (3), PB (3 + expectancy multiplier), and TV (3) each require participant-specific random effects on the maxima.
- At trial-relevant N, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts.
- Additional parametric structure typically inflates moderation estimand variance more than it reduces bias.

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The second reason is degrees of freedom. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation programme) is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz BR (three parameters), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

[57]

The third constraint is convergence fragility: component-matched nonlinear mixed models converge below 90% at trial-relevant N, contaminating aggregated analyses.

- Simultaneous Gompertz BR, PB, and TV fitting requires `nlme::nlme`, Bayesian hierarchical, or structural-equation approximations.
- Each is more sensitive to starting values, prone to local optima, and likely to fail on individual trial datasets than the LME analogue.
- Non-converging cells at below-90% convergence rates contaminate power and bias estimates in simulation studies.

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The third reason is convergence fragility. Component-matched analyses are non-linear in the parameters and their convergence is problematic in practice. Fitting Gompertz BR, PB, and TV simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant N, the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

[58]

The fourth constraint is estimand robustness: the `bm:Dbc` marginal interaction is the right estimand for the biomarker-validation question regardless of the BR-PB split.

- Regulators want to know whether the biomarker predicts treatment response under trial conditions, not a mechanistically decomposed prediction.
- The marginal `bm:Dbc` interaction captures this question whether the moderation is biological or expectation-mediated.
- Component decomposition adds mechanistic precision but is not required for the core biomarker-validation decision.

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The fourth reason is that the moderation estimand is sufficiently robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

[59]

The inferential target is stated as an estimand: the population biomarker-by-treatment interaction with explicit intercurrent-event handling, while the pharmacological-component target is more ambitious and inherits the heterogeneity-identifiability caveat.

- The $bm{:}D_{bc}$ estimand is defined by its population, treatment contrast, endpoint, summary, and intercurrent-event strategy in the ICH E9(R1) sense.
- Dropout is handled by a treatment-policy strategy and residual carryover by the modelled exponential decay.
- Recovering a participant-level pharmacological interaction inherits Senn's caveat that true heterogeneity is separable from within-person variation only under repeated phase contrasts.

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It is worth stating the inferential target of the analysis in the vocabulary of the estimands framework[53], because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-by-treatment interaction, the $bm{:}D_{bc}$ coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarised through the continuous indicator D_{bc}, its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modelled exponential decay for residual carryover. The pharmacological-component target β_{bm}^{BR} is a more ambitious estimand: it is the biomarker-by-BR interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasised for individual treatment effects[49,54]: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

5.2 A small extension that usually pays off: phase indicators

[60]

The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.

- `phase` absorbs systematic differences in PB expectancy across trial phases.
- `Dbc:phase` detects whether the drug effect shrinks under blinding, a signature of PB-mediation.
- `bm:Dbc:phase` detects whether the biomarker-by-treatment interaction shrinks under blinding, a signature that it is partly biomarker-by-PB.

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The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
       + bm:Dbc + bm:Dbc:phase
       + (1 | ptID),  correlation = corCAR1(...)
```

Here `phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR.

[61]

The phase-augmented extension adds at most four parameters yet captures the practically relevant belief-attenuation diagnostic without requiring the Gompertz parametric form.

- Three to four parameters is well within budget at any realistic trial sample size.
- The extension does not separately estimate BR and PB components but identifies the change in apparent drug effect when belief is hedged.

- For most analyses this is the practically relevant question and is answerable without committing to the Gompertz form.

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This extension adds three to four parameters at most, well within budget at any realistic N. It does not separately estimate BR and PB as components, but it does identify the change in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form. Where both the lumped and the phase-augmented analyses are reported, the additional `Dbc:phase` and `bm:Dbc:phase` terms introduce a small family of belief-attenuation tests, and the attribution inference they support should be treated as a pre-specified secondary analysis with its multiplicity acknowledged rather than as a second bite at the primary interaction test.

5.3 When to fit the full decomposition

[62]

The full decomposition merits the effort only when N is in the hundreds, the design is fully phase-balanced, and the scientific question requires the BR-PB mechanistic split.

- Feasibility conditions: several-hundred N, dense timepoints in every phase, mechanistic BR-PB split required, and informative Bayesian priors available.
- In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or `brms` can return participant-specific component estimates.
- Outside that regime the phase-augmented LME is the better tool.

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A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

[63]

The DGP-analysis asymmetry is deliberate: simulation uses the rich DGP to characterise how a simpler analysis behaves; the analysis need not mirror the DGP to produce useful inference.

- Simulation controls the DGP and can specify any structure; inference must pay the identifiability cost of every parameter.
- The DGP is a characterisation tool; the analysis is a finite-data inference tool; they have different optimality criteria.
- DGP-matching is not required and is often counterproductive at trial-relevant sample sizes.

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The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterise how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

[64]

The exponential-decay carryover indicator is a deliberately parsimonious special case of richer carryover models, and the simulation should benchmark it against a distributed-lag alternative.

- The continuous indicator D_{bc} decays as $\exp(-\lambda t_{sd})$, a one-parameter form tied to a pharmacokinetic half-life.
- Distributed-lag models, the average-period-treatment-effect estimand, and G-estimation relax the single-rate assumption at additional parametric cost.
- The exponential restriction is an assumption, not a finding, and the pre-registered simulation should include a distributed-lag arm to quantify its cost.

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A word on the carryover parameterisation is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator D_{bc} decays as $\exp(-\lambda t_{sd})$ off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al.[31] model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional parameters; Daza[55] formalises an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and G{"a}rtner et al.[56] benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. The pre-registered simulation programme of section 7 should accordingly include a distributed-lag analysis arm, so that the cost of the exponential-decay restriction is quantified against the richer alternative rather than asserted.

6 Worked example: information loss without decomposition

6.1 What the lumped analysis estimates

[65]

The lumped treatment-effect estimate has a probability limit equal to the pharmacological component plus belief and natural-history contributions, so its bias does not vanish as N grows.

- Writing the reduction as $BR + PB + TV - \varepsilon$ and fitting one drug indicator gives plim $\hat{\beta}_D = m_{BR} + b_{PB} + b_{TV}$.
- b_{PB} and b_{TV} are the belief and natural-history terms that load onto the drug indicator under the chosen contrast; both are generally non-negative for an improving outcome.
- Because the displacement is in the probability limit, a larger one-component trial converges on the same biased value: the failure is identification, not precision.

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Before the numerical illustration it is worth establishing algebraically what the one-component analysis actually estimates, because the qualitative failure modes of section 1 are consequences of an identity rather than empirical tendencies that require simulation to demonstrate. Write the within-participant symptom reduction as

$R_{it} = \mathrm{BL}_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$. The one-component analysis fits a single drug indicator D_{it},

$$R_{it} = \alpha + \beta_D D_{it} + g(t) + u_i + e_{it},$$

and reports $\hat{\beta}_D$ as 'the treatment effect'. Taking the on-drug minus off-drug contrast that β_D targets, and using that the natural-history component is by construction independent of treatment status so that its on/off difference cancels in a within-participant comparison, the probability limit of the estimator is

$$\operatorname{plim} \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV},$$

where $b_{PB} = E[PB \mid \text{on}] - E[PB \mid \text{off}]$ is the belief differential between the contrasted phases and b_{TV} collects any natural-history contribution that fails to cancel under the contrast actually used; b_{TV} is zero for a clean on/off within-participant contrast and strictly positive for a baseline-anchored pre-post analysis, in which case $b_{PB} = E[PB]$ and $b_{TV} = E[TV]$. To fix which case is in play below: the one-component analysis used in the worked example and the simulations of this paper is the design-contrast (drug-on/off) form, so $b_{TV} \approx 0$ there and the operative displacement is the belief term b_{PB}; the baseline-anchored pre-post form, which additionally carries $b_{TV} = E[TV]$, is the worse case retained only for comparison. Both terms are generally non-negative for an improving outcome, so the lumped estimate over-states the pharmacological component m_{BR}. The displacement is a property of the probability limit and is unchanged as $N \to \infty$: a larger one-component trial converges on the same biased value. This is the precise sense in which the placebo-attribution failure of section 1 is a problem of identification, not of precision.

[66]

The biomarker slope from a lumped analysis decomposes additively into component-specific slopes, so a biomarker correlated with PB or TV registers as a predictor even with no pharmacological association.

- By linearity of covariance, $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$.
- A non-zero β_{bm}^{PB} or β_{bm}^{TV} produces a non-zero lumped slope even when the pharmacological slope β_{bm}^{BR} is exactly zero.
- The result is an algebraic identity under the additive decomposition, not a sample-size-dependent tendency.

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The same algebra settles the biomarker-validation failure. The precision-medicine analysis regresses a participant-level response summary on a baseline biomarker B_i. If that summary is the lumped per-participant effect, which estimates $m_{BR,i} + w_{PB}\, m_{PB,i} + w_{TV}\, m_{TV,i}$ for design-determined loadings $w_{PB}, w_{TV} \geq 0$, then by the linearity of covariance the fitted biomarker slope has probability limit

$$\beta_{bm}^{\text{lumped}} \;=\; \frac{\operatorname{Cov}(m_{BR} + w_{PB} m_{PB} + w_{TV} m_{TV},\, B)}{\operatorname{Var}(B)} \;=\; \beta_{bm}^{BR} + w_{PB}\, \beta_{bm}^{PB} + w_{TV}\, \beta_{bm}^{TV},$$

where $\beta_{bm}^{C} = \operatorname{Cov}(m_C, B)/\operatorname{Var}(B)$ is the component-specific slope. A biomarker correlated with placebo responsiveness or with the natural-history trajectory, so that β_{bm}^{PB} or β_{bm}^{TV} is non-zero, therefore registers a non-zero lumped slope even when its pharmacological association β_{bm}^{BR} is exactly zero. This is the biomarker-validation failure of section 1, and it too is an identity under the additive decomposition rather than an empirical tendency: no sample size separates the lumped slope from a genuine pharmacological one, because the two coincide in the estimand whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$.

We make two qualifications explicit, because they bound how much weight the framework can carry. First, both displays are immediate corollaries of the additive decomposition and the linearity of the projection that defines the estimator; they are not a derived result in any deeper sense, and their value is in naming the displacement terms, not in the algebra. Second, they assume the components combine additively. Under the subadditive interaction documented in section 3.3, where the joint effect of BR and PB falls below their sum, a first-order correction proportional to $\operatorname{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the biomarker slope, and the identities above should be read as the leading behaviour in the additive case rather than as exact under interaction. The empirical magnitude of that correction term is itself an open question that the framework's γ-interaction simulation (section 3.3) is designed to address.

[67]

The bias displaces the estimand rather than inflating variance, so a significance test of the lumped coefficient cannot detect it; the decomposition is what locates it.

- The lumped coefficient is displaced from m_{BR} by $b_{PB} + b_{TV}$ but remains a well-estimated quantity with a valid Wald test of its own nullity.
- Detecting the bias requires comparing $\hat{\beta}_D$ to m_{BR}, a comparison only the decomposition or an isolating phase contrast supplies.

- This is why a rejection-rate summary is the wrong instrument, and accounts for the flat rejection rates across the placebo-strength axis in the section 7.1 pilot.

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Two consequences for how the framework should be evaluated follow. First, the quantity that carries the bias is the location of the estimand, $\operatorname{plim}\hat{\beta}_D$, not its sampling variability: the lumped coefficient remains a well-estimated number with a valid Wald test of its own nullity, and that test rejects at the rate dictated by the displaced value. A significance test of the lumped coefficient therefore cannot, in principle, detect the bias, because the bias does not move the coefficient toward zero. Detecting it requires comparing $\hat{\beta}_D$ against m_{BR}, which is exactly the comparison the decomposition, or a phase contrast that isolates m_{BR}, makes available and the lumped analysis does not. Second, and as a corollary for the simulation evidence, a rejection-rate or power summary is the wrong instrument for this bias: the right target is the coefficient itself and its offset from m_{BR}. This is why the pilot of section 7.1 found the biomarker-by-treatment rejection rate essentially flat across the placebo-strength axis even though that axis is precisely what drives the predicted bias; the prediction lives in the mean of $\hat{\beta}_{bm:D_{bc}}$, and a production run that means to test it must read the bias off the coefficient, not off the test.

6.2 A numerical illustration

[68]

The worked example uses two participants with identical lumped responses but opposite BR-PB profiles to show that the one-component analysis cannot distinguish them.

- Participant A has high PB ($m_{PB} = 6$) and moderate BR ($m_{BR} = 4$); Participant B has low PB ($m_{PB} = 1$) and high BR ($m_{BR} = 8$).
- Both participants reach approximately 10–11 points of total improvement by week 8, making them indistinguishable to a one-component analysis.
- The blinded-discontinuation phase creates the diagnostic contrast that exposes the underlying difference.

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To illustrate the inferential cost of failing to decompose, we shall work through a simple numerical example. Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly. Participant A is a high placebo responder and a moderate pharmacological responder, with true component values $m^A_{BR} = 4$, $m^A_{PB} = 6$, $m^A_{TV} = 1$. Participant B is a low placebo responder and a strong pharmacological responder, with $m^B_{BR} = 8$, $m^B_{PB} = 1$, $m^B_{TV} = 1$.

[69]

Open-label totals are clinically indistinguishable; the one-component analysis estimates a population drug effect of approximately 10 points with no differentiation between A and B.

- Both participants show approximately 10–11 points of improvement and present as clinical successes.
- A t-test or simple LME on open-label change cannot detect the heterogeneity in BR versus PB composition.
- The population drug-effect estimate conflates both pharmacological and placebo contributions.

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In the open-label phase (weeks 1 to 8), both BR and PB run at full strength and TV accumulates. By week 8 both participants have reached close to their saturation values:

	BR	$PB(\eta = 1)$	TV	Total improvement
Participant A	4.0	6.0	1.0	**11.0**
Participant B	8.0	1.0	1.0	**10.0**

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

[70]

The blinded-discontinuation phase reveals the BR-PB heterogeneity: Participant A retains 4 points while B collapses to 1.5, exposing the diagnostic contrast.

- Participant A's high PB at $\eta = 0.5$ (3 points) sustains improvement even without active drug.
- Participant B's low PB (0.5 points) and absent BR produce near-complete return to baseline.

- The 4.0 versus 1.5 contrast is exactly what the decomposition uses to separate BR from PB.

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In the blinded discontinuation phase (weeks 9 to 12), both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. For clarity of exposition we take the carryover half-life to be short relative to the four-week blinded window, so that BR has effectively cleared by the end of week 12; the more realistic case of a slowly decaying BR, modelled by the carryover machinery of section 3.1, attenuates but does not eliminate the contrast below. By the end of week 12, the components are:

	BR	$PB(\eta = 0.5)$	TV	Total improvement
Participant A	0.0	3.0	1.0	**4.0**
Participant B	0.0	0.5	1.0	**1.5**

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits. For simplicity the table holds TV at its week-8 value; in reality TV continues to accumulate over weeks 9 to 12, which shifts both totals by a common amount and leaves the BR-versus-PB contrast on which the example turns unchanged.

[71]

The one-component model estimates drug at 7.75 points with no principled decomposition, obscuring that A's true BR is 4 and B's is 8.

- The 7.75 point estimate is a conflated average of true BR values (4.0 and 8.0) plus the placebo wash-out.
- The model cannot determine whether A and B differ, nor estimate the pharmacological drug effect separately.
- Placebo and natural-history effects are not identifiable from this contrast alone.

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A one-component fixed-effects regression of total symptom change on phase-by-allocation indicators sees:

- Open-label means: 11.0 (A), 10.0 (B).
- Blinded-placebo means: 4.0 (A), 1.5 (B).

- Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B). Population mean: 7.75.
- Estimated placebo effect: not identifiable from this contrast.
- Estimated natural history: not identifiable from this contrast.

The one-component model has produced a single number conflated with the placebo response that washes out under blinding. It cannot say whether the drug effect is the same for participants A and B; it estimates 'drug' at 7.75 points on average, which is an average of the true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, with no principled way to decompose further.

[72]

The full decomposition recovers all six component estimates by exploiting phase differential expectancy and the blinded-placebo contrast.

- The LME returns $\hat{m}^A_{BR} = 4.0$, $\hat{m}^B_{BR} = 8.0$, $\hat{m}^A_{PB} = 6.0$, $\hat{m}^B_{PB} = 1.0$ at the population point estimate.
- Participant-level estimates have meaningful uncertainty at trial-relevant N; the simulation study characterises this.
- The phase differential ($\eta = 1$ vs. $\eta = 0.5$) is the identification mechanism for the PB parameter.

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A linear-mixed-effects analysis with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. After fitting, the model returns $\hat{m}^A_{BR} = 4.0$, $\hat{m}^B_{BR} = 8.0$, $\hat{m}^A_{PB} = 6.0$, $\hat{m}^B_{PB} = 1.0$, $\hat{m}^A_{TV} = \hat{m}^B_{TV} = 1.0$ (at the population point estimate; participant-level estimates have meaningful uncertainty at N in the trial-relevant range, which the simulation study below characterises).

[73]

The one-component model produces one biased average; the three-component model produces six clinically actionable quantities, making biomarker validation possible.

- The lumped estimate of 7.75 points is a biased average of $BR_A = 4$ and $BR_B = 8$ confounded with placebo wash-out.
- The decomposition supplies individual BR, PB, and TV estimates enabling per-participant clinical decisions.
- Predictive-biomarker validation requires the decomposition because biomarker-by-BR is the target estimand.

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The information loss from lumping the components is not abstract. We may compare the inferences directly:

Question	One-component answer	Three-component answer
Does the drug work for A?	~7.75 points (lumped).	$BR_A = 4.0$, distinct from her larger placebo response.
Does the drug work for B?	~7.75 points (lumped).	$BR_B = 8.0$, twice the magnitude of A.
Are A and B different responders?	No discernible difference.	B has 2x A's BR but 1/6 of A's PB.
Is a biomarker predicting BR useful here?	Cannot answer.	Yes if the biomarker correlates with $\hat{m}_{BR}$.
Should A be continued long-term?	Cannot say.	Probably yes ($BR = 4$ is real and pharmacological), but the larger placebo component will not persist if expectations decay.
Population mean drug effect?	7.75 points (biased).	$6.0 = (4.0 + 8.0)/2$ (correct mean of BR).

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. Predictive-biomarker validation, which is a central scientific aim of the trial design family considered here, becomes impossible without the decomposition.

[74]

A population-level example shows that cohort differences in open-label totals driven entirely by PB produce apparent non-replicability that the decomposition resolves.

- Cohort 1 (naive, $m_{PB} = 7$) reports 13-point improvements; Cohort 2 (sceptical, $m_{PB} = 2$) reports 8 points; true $m_{BR} = 5$ in both.
- A one-component analysis would falsely conclude the drug works less well in Cohort 2 and trigger searches for confounders.

- The decomposition returns $\hat{m}_{BR} = 5$ in both cohorts, attributing the difference correctly to PB heterogeneity.

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A second example at the population level illustrates the same problem. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug. Cohort 1 (early trial, naive patients) has mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$. Cohort 2 (later trial, patients who have had multiple prior failed treatments and are sceptical) has mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$. By construction the drug works equally well in both cohorts. But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug works less well in cohort 2 and might trigger a search for confounders or sub-population effects. None of this is real; the difference is entirely in PB, specifically in the population baseline level of placebo responsiveness. The three-component decomposition, applied to either cohort, returns $\hat{m}_{BR} = 5$ in both. Unfortunately, this is the kind of apparent non-replicability that pharmacology has spent the last decade struggling to navigate; the decomposition is one of the formal tools that makes it tractable.

7 Simulation study

[75]

The simulation study follows ADEMP and quantifies bias and identifiability properties of three analysis strategies across a prazosin-PTSD-calibrated parameter grid.

- The Morris (2019) ADEMP framework structures the study: Aims, DGP, Estimands, Methods, Performance measures.
- The pre-registration is at `analysis/scripts/component-decomposition/00-ademp-pre-registrat`
- Three analysis strategies (one-component, phase-augmented, full decomposition) are compared across the parameter grid.

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The simulation study reported and pre-registered in this section is structured under the[57] ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The pre-registered plan is at `analysis/scripts/component-decomposition/00-ademp-pre-registration.md`. A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in section 5 across a parameter grid.

7.1 Pilot validation of analysis machinery (100 replicates, 6 cells)

[76]

The pilot is a structural check of the analysis pipeline, not a substantive empirical result.

- The realised Study A run (section 7.3) is the empirical core of the manuscript; this pilot precedes it.
- The pilot confirms only that data generation, model fitting, and summary output run without error.
- No bias or power claims are derived from the 100-replicate pilot.

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The production simulation programme is the empirical core of this manuscript; its first realised run, Study A, is reported in section 7.3, while the fuller pre-registered grid (section 7.2) remains partly outstanding. The pilot reported here precedes both and confirms only that the analysis machinery runs end-to-end.

[77]

The pilot executed a 6-cell, 100-replicate run with perfect convergence, confirming the pipeline is operational.

- The grid crossed three *pb_strength* levels against two analysis strategies.
- Convergence was 1.0 across all six cells under 8-core `mclapply`.
- Output is checkpointed at `analysis/data/quick-sim/`; no inferential claims are drawn.

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A 100-replicate-per-cell pilot was run at the reference Hybrid OL+BDC design, $N = 35$, on the 6-cell slice $pb\_strength \in \{0, 6.5, 13\} \times$ analysis $\in$ {one-component, phase-augmented}, using 8-core `parallel::mclapply`. The convergence rate was 1.0 across all six cells. Per-replicate output is checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds` and the Morris-format cell-level summary at `analysis/data/quick-sim/06-decomp-summary.txt`. The exploratory summaries are diagnostic only. The one-component `bm:Dbc` test showed power 0.95, 0.93, and 0.83 across $pb\_strength \in \{0, 6.5, 13\}$, while the phase-augmented analysis showed 0.65, 0.65, and 0.61; the mean estimated $\hat{\beta}_{bm:D_{bc}}$ was essentially flat within each analysis (about -2.25 for the one-component and −2.21 for the phase-augmented fit). At 100 replicates the Monte Carlo standard error on a power estimate near 0.7 is about 0.046, so these figures cannot by themselves separate the two analyses; their interpretation, including why the phase-augmented analysis loses its bias-detection contrast at this N, is resolved by the production run of section 7.3. No confirmatory bias or power claim is drawn from the pilot, and all substantive claims are deferred to the production programme below.

[78]

Two pilot findings revise the production specification: rank-deficiency forces a formula change, and bias must be read from coefficient means, not rejection rates.

- At $N = 35$, rank-deficiency dropped the D_{bc}:phase term, eliminating the key biomarker-by-treatment-by-phase contrast.
- The predicted bias manifests in the mean of $\hat{\beta}_{bm:D_{bc}}$, not in the Wald-test rejection rate.
- The production programme addresses both by using larger N, a fuller design, and coefficient-mean reporting.

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Two design-level observations from the pilot inform the production specification. First, at $N = 35$ the rank-deficiency of the full phase-augmented formula required dropping the D_{bc}:phase interaction term, removing the precise contrast that would have isolated the biomarker-by-treatment-by-phase signal the framework is designed to detect. Second, the predicted bias of the one-component analysis arises in the regression coefficient $\hat{\beta}_{bm:D_{bc}}$, not in the indicator-level rejection rate, so power on the biomarker-by-treatment Wald test is not the right summary statistic to detect the framework's predicted bias. The production programme below addresses both observations through larger N, a richer trial design that admits the full phase-augmented formula, and reporting of the coefficient mean against the implied true value rather than the rejection rate alone.

7.2 Production-simulation specification

[79]

The production programme follows the ADEMP framework with four deliverables; the realised Study A run in section 7.3 is the first of these.

- The deliverables are interaction-estimate bias, full-decomposition identifiability, type I error and power, and participant-level signal recovery.

- Sample sizes $N \in \{35, 70, 100, 150\}$ cross a *pb_strength* grid; type I cells target MCSE 0.003 at 5,000 replicates.
- Driver scripts, checkpointed outputs, and the ADEMP pre-registration reside under `analysis/scripts/component-decomposition/`.

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The production programme follows the[57] ADEMP framework and targets four deliverables: the bias of $\hat{\beta}_{bm:D_{bc}}$ for the one-component, phase-augmented, and full-decomposition analyses across $N \in \{35, 70, 100, 150\}$ and a *pb_strength* grid; identifiability metrics for the full-decomposition fit (convergence, variance-inflation factors, and sensitivity of the BR estimate to the η(phase) multipliers); type I error and power for the interaction test under an η-mismatch, TV-magnitude, and dropout sensitivity grid, with null cells at $n_{\text{reps}} = 5{,}000$ for MCSE 0.003; and the participant-level interaction signal recovered as a function of (m_{BR}, m_{PB}, m_{TV}), which the lumped-total regression does not preserve. The `tidyverse` implementation collection drives the grid; driver scripts, checkpointed cell-level summaries, per-replicate output, and the committed ADEMP pre-registration all reside under `analysis/scripts/component-decomposition/`. The realised Study A run reported next used 1,000 replicates per alternative cell.

7.3 Study A results: bias and power at 1,000 replicates per alternative cell

[80]

At 1,000 replicates per alternative cell the one-component estimate of $\hat{\beta}_{bm:D_{bc}}$ is essentially unbiased across the entire (m_{PB}, m_{TV}) grid, contradicting the framework's prediction of a placebo-strength-driven bias.

- The true biomarker-by-BR effect is -2.25; mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one Monte Carlo SE of zero.
- Bias shows no monotone trend across $m_{PB} \in \{0, 1, 3, 6, 10\}$ or $m_{TV} \in \{-1, 0, 1, 2\}$; it behaves as sampling noise around zero.
- This is the predicted consequence of the identity in section 6.1: the DGP couples the biomarker to BR only, so $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$ and the lumped slope is unbiased by construction.

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The production Study A run (1,000 replicates per alternative cell, 5,000 per null cell, 80 alternative cells crossing $m_{PB} \in \{0, 1, 3, 6, 10\}$,

$m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 100, 150\}$, plus four null cells) returns a result that does not support the framework's central bias prediction. The data-generating process couples the baseline biomarker to the BR component only, through the on-drug biomarker-BR covariance channel, so the true biomarker-by-treatment effect is the biomarker-by-BR slope, $\beta_{bm:D_{bc}} = -c_{bm}\,\sigma_{BR}/\sigma_{bm} = -2.25$, and the biomarker is by construction uncorrelated with PB and TV. Under the one-component analysis the mean estimated $\hat{\beta}_{bm:D_{bc}}$ tracks this value across the whole grid: the mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one Monte Carlo standard error (respectively about 0.028, 0.019, 0.009, and 0.007) of zero, with no monotone trend across m_{PB} or m_{TV}. The predicted placebo-attribution bias of the biomarker-by-treatment estimate does not appear.

This is not a failure of the simulation; it is the behaviour the identity of section 6.1 requires. That identity writes the lumped biomarker slope as $\beta^{BR}_{bm} + w_{PB}\beta^{PB}_{bm} + w_{TV}\beta^{TV}_{bm}$. With the biomarker coupled to BR alone, $\beta^{PB}_{bm} = \beta^{TV}_{bm} = 0$, so the lumped slope equals the pharmacological slope no matter how large m_{PB} or m_{TV} grows. Study A varies the magnitudes of the unmodelled components, which the identity shows to be the innocuous axis; the axis that produces biomarker-validation bias is the biomarker's correlation with PB or TV, which this DGP holds at zero. The clean null is therefore a confirmation of the identity and a correction to the looser claim of section 1 that large PB or TV by itself biases biomarker validation. It does not; biomarker-component coupling does.

[81]

The phase-augmented analysis provides no bias reduction and incurs a substantial power penalty, so it is dominated under this uncontaminated DGP.

- Mean power at $N = 35$ is 0.35 (phase-augmented) versus 0.59 (one-component); at $N = 70$, 0.62 versus 0.90; both reach approximately 1.0 by $N = 100$.
- Phase-augmented mean absolute bias is no smaller than one-component at any N (0.028 versus 0.016 at $N = 35$).
- Convergence is approximately 100% for both analyses at every N (minimum 0.998), so the rank-deficiency that degraded the $N = 35$ pilot does not recur here.

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The comparison between analyses is equally clear. The phase-augmented analysis is not less biased than the one-component analysis at any sample size (its mean absolute bias is 0.028 versus

0.016 at $N = 35$ and is comparable elsewhere), and it is markedly less powerful for the biomarker-by-treatment test: mean power across the grid is 0.35 versus 0.59 at $N = 35$ and 0.62 versus 0.90 at $N = 70$, with both analyses reaching essentially full power by $N = 100$. Because the DGP contains no biomarker-PB contamination for the phase contrast to remove, the extra phase-by-treatment terms add estimation cost without buying any bias reduction, and the analysis is dominated. Convergence was approximately complete for both analyses at every sample size (minimum convergence rate 0.998), so the rank-deficiency that forced dropping the `Dbc:phase` term in the $N = 35$ pilot, which we had advanced as the explanation for the pilot's inverted power ranking, is not operative in the production run: the inverted ranking persists with the full formula fitted and converged. The pilot's power inversion was therefore a genuine feature of this DGP, not an artifact of rank-deficiency.

Calibration is adequate for both analyses. Nominal-95% confidence-interval coverage of $\beta_{bm:D_{bc}}$ ranges from 0.96 to 0.99 for the one-component analysis (slightly conservative) and from 0.94 to 0.98 for the phase-augmented analysis. Type I error at the four null cells is controlled, ranging from 0.031 to 0.056 for the one-component analysis and 0.038 to 0.049 for the phase-augmented analysis; the one-component estimate at $N = 100$ (0.056, MCSE 0.003) sits 1.9 standard errors above nominal but is within the simulation's Monte Carlo uncertainty.

[82]

Study A confirms the identity but cannot exhibit the biomarker-validation failure; a contaminated-biomarker pilot reported in the next subsection provides the decisive evidence.

- As pre-registered, Study A varies component magnitude with the biomarker fixed to BR, so it probes an axis the identity predicts to be null.
- The decisive axis is $c_{bm,PB}$, the biomarker-PB correlation; this is varied in the contaminated-biomarker pilot below.
- On current evidence the phase-augmented analysis is not recommended for the uncontaminated-biomarker regime; the pilot below evaluates it under contamination.

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Two conclusions follow for the Study A programme. First, the result sharpens rather than supports the section 1 framing: the biomarker-validation failure is real as an identity, but it is driven by the biomarker's coupling to PB or TV, not by the magnitude of those components, and Study A as pre-registered varies the latter while holding the former at zero. The decisive axis is the biomarker-PB correlation $c_{bm,PB}$, which the contaminated-biomarker pilot in the following subsection varies directly. Second, on the present evidence the phase-augmented analysis carries a real power cost without a compensating bias reduction under an uncontaminated biomarker, so it should not be recommended as a default for this design; the pilot below evaluates whether contamination changes this verdict.

7.4 Contaminated-biomarker pilot (100 replicates per cell)

[83]

A pilot varying the biomarker-PB correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ confirms that bias in the one-component estimator scales with coupling strength, not component magnitude, and that the phase-augmented analysis provides no protection.

- At $N = 150$, mean $\hat{\beta}_{bm:D_{bc}}$ shifts from -2.25 (true) to -1.74 at $c_{bm,PB} = 0.45$ (bias $+0.51$, 30 Monte Carlo SEs from zero), confirming the identity's prediction.
- The bias is driven by $c_{bm,PB} \times \sigma_{PB}$, not by the population-mean m_{PB}: cells with $m_{PB} = 0$ and $m_{PB} = 6$ show identical bias at every $c_{bm,PB}$ level, because the individual-level PB variance ($\sigma_{PB} = 2$) provides the contamination channel even when the mean is zero.
- The phase-augmented analysis shows the same bias as the one-component analysis at every contamination level, offering no protection against biomarker-PB coupling.
- This pilot uses 100 replicates per cell; a production run at 1,000 replicates per cell is required to confirm the pattern and assess type I error under contamination.

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A supplementary pilot varying the biomarker-PB correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ at two PB magnitude levels ($m_{PB} \in \{0, 6\}$) and two sample sizes ($N \in \{70, 150\}$), using 100 replicates per cell, confirms the identity-based prediction and reveals an unexpected structural result.

At $N = 150$ the one-component bias scales monotonically with $c_{bm,PB}$: mean $\hat{\beta}_{bm:D_{bc}}$ moves from -2.31 ($c_{bm,PB} = 0$, bias -0.06) to -2.37 ($c_{bm,PB} = 0.15$, bias -0.12) to -2.08 ($c_{bm,PB} = 0.30$, bias $+0.17$) to -1.74 ($c_{bm,PB} = 0.45$, bias $+0.51$). The last value sits 30 Monte Carlo standard errors from the true -2.25, establishing the effect unambiguously. The direction of bias is positive (toward zero): high-$c_{bm,PB}$ participants experience stronger PB, which in the open-label phase adds to the apparent drug effect, but this elevation is partially absorbed by the bm main effect, leaving the $bm : D_{bc}$ interaction coefficient attenuated relative to the pure pharmacological value.

The unexpected result is that m_{PB} does not modulate the bias: at the contaminated cells ($c_{bm,PB} \geq 0.30$, $N = 150$) the $m_{PB} = 0$ and $m_{PB} = 6$ biases are identical to within 0.006, well inside Monte Carlo error, even as the bias itself grows to +0.51; the only larger discrepancy is at $c_{bm,PB} = 0$, where both biases are near zero and differ by 0.067 at the pilot's 100-replicate resolution. The driver is therefore $c_{bm,PB} \times \sigma_{PB}$, not $c_{bm,PB} \times m_{PB}$. Even when the population-mean PB amplitude is zero, individual participants have nonzero PB draws (standard deviation $\sigma_{PB} = 2$), and coupling the biomarker to that variance is sufficient to contaminate the estimator. A true no-PB control requires $\sigma_{PB} = 0$, not $m_{PB} = 0$. This refines the Study A design: varying m_{PB} at $c_{bm,PB} = 0$ correctly produced zero bias, but because the relevant null was $c_{bm,PB} = 0$, not $m_{PB} = 0$.

The phase-augmented analysis provides no protection: its bias at every $(c_{bm,PB}, m_{PB}, N)$ cell is indistinguishable from the one-component bias. Adding phase-by-treatment interactions to the model does not decompose the contamination because both analyses see the biomarker-PB covariance through the same open-label versus blinded-discontinuation contrast; the contrast changes the weight on PB (via expectancy) but does not isolate the PB component's contribution to the biomarker slope. Separating that contribution requires both a design that decouples drug exposure from belief and a decomposition that exploits the decoupling; Study B (section 7.5) shows that the standard hybrid design cannot achieve this separation, whereas a balanced-placebo design can.

The $N = 70$ results are uninformative at 100 replicates (Monte Carlo standard error ≈ 0.06, comparable to or larger than the bias at moderate $c_{bm,PB}$). The production run will use 1,000 replicates per cell to establish the $N = 70$ pattern and 5,000 null replicates to assess type I error under contamination. Until the production run is complete, the quantitative bias estimates from this pilot should be treated as directionally informative but imprecise.

[84]

The contaminated-biomarker pilot reframes the paper's practical guidance: the risk of biomarker-validation failure arises from biomarker-PB correlation in the study population, not from the presence of strong placebo response per se.

- A trial in a population where the biomarker is uncorrelated with placebo responsiveness can use the one-component estimator without bias, regardless of how large the placebo response is.

- A trial where the biomarker covaries with optimism, suggestibility, or treatment-seeking history is at material risk of contamination; bias of +0.51 (23% of the true effect) at $c_{bm,PB} = 0.45$ illustrates the practical scale.
- Neither the one-component nor the phase-augmented analysis removes this bias; full component decomposition is required.

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The pilot results reframe the paper's practical guidance. The risk of biomarker-validation failure in an aggregated N-of-1 study arises not from the presence of a strong placebo response in the population but from the biomarker's correlation with individual-level placebo responsiveness. In a population where the baseline biomarker is uncorrelated with each participant's placebo-belief response amplitude, the one-component estimator recovers the pharmacological interaction without bias, and the full decomposition machinery is not required. In a population where the biomarker covaries with psychological characteristics that predict placebo response – optimism, prior treatment experience, suggestibility, expectancy at enrolment – the one-component analysis conflates pharmacological and placebo-belief prediction, and the scale of that conflation grows with the coupling strength. At $c_{bm,PB} = 0.45$, a value that equals the pharmacological coupling used throughout this paper, the one-component bias is approximately 0.51 points on the interaction scale, representing 23% of the true pharmacological effect. This is a practically meaningful distortion for any precision-medicine stratification decision.

The production run of the contaminated-biomarker study, at 1,000 alternative-cell replicates and 5,000 null-cell replicates, is required to confirm these estimates, establish the $N = 70$ pattern, and characterise type I error under contamination. Results are pending.

7.5 Study B: recovery under a belief-decoupling design

[85]

Study B shows the contamination is removable, but only under a design that decouples drug exposure from belief: on a balanced-placebo design a belief-covariate decomposition recovers the unbiased pharmacological slope across the contamination range, while the one-component estimator does not.

- On the standard hybrid design the drug state D_{bc} and the expectancy η are collinear, so no decomposition can separate the channels; we confirmed that several tractable decompositions fail there, no better than the one-component analysis.

- A balanced-placebo design adds a covert on-drug phase (drug given, belief withheld, $\eta = 0$) and an open-placebo phase (drug withheld, belief sustained, $\eta = 1$), which decouples D_{bc} from η.
- The decomposition $Sx \sim bm + t + D_{bc} + \eta + bm{:}D_{bc} + bm{:}\eta$ then isolates β_{bm}^{BR} on the $bm{:}D_{bc}$ term while $bm{:}\eta$ absorbs the placebo channel.

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Study B asks whether the contamination demonstrated in section 7.4 can be removed by decomposition. The answer is conditional on the trial design. The mechanism is that, in the MVN data-generating process, the biomarker-BR coupling acts only on drug (and so loads on the drug-state contrast D_{bc}), whereas the biomarker-PB coupling scales with the expectancy η (the placebo component's amplitude is proportional to η, so its biomarker-coupled part loads on η). A decomposition that interacts the biomarker with both D_{bc} and η can therefore isolate β_{bm}^{BR} on $bm{:}D_{bc}$, but only if D_{bc} and η vary independently in the design. In the standard hybrid open-label-plus-blinded-discontinuation design they do not: on-drug time and belief accumulate together (their analysis-stage correlation is about +0.6), and we verified that the belief-covariate decomposition, a blinded-stratum drug contrast, and a Gompertz-basis component decomposition all fail to recover β_{bm}^{BR} on that design, performing no better than, and sometimes worse than, the one-component analysis. The remedy is a design that breaks the collinearity. A balanced-placebo (open-hidden) design[51,52] interposes a covert on-drug phase, in which the drug is administered without the participant's knowledge ($\eta = 0$), and an open-placebo phase, in which the participant is off drug but believes treatment continues ($\eta = 1$); the first supplies drug variation at fixed belief and the second belief variation at fixed drug exposure, reducing the D_{bc}-η correlation to about −0.66.

[86]

At 1,000 replicates per cell on the balanced-placebo design, the decomposition is essentially unbiased across the contamination range while the one-component estimator degrades monotonically.

- At $N = 150$, the decomposition's bias of $\hat{\beta}_{bm:D_{bc}}$ stays within one Monte Carlo SE of zero across $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ (−0.022, +0.008, +0.021, −0.026; MCSE ≈ 0.02), while the one-component bias rises monotonically to +0.16, +0.34, +0.48.
- Nominal-95% coverage is near-nominal for the decomposition

(0.94–0.96) and falls to 0.86 for the one-component analysis at the highest contamination; convergence is 100 percent for both.

- The contamination grid is capped at $c_{bm,PB} = 0.30$ because beyond approximately 0.45 the implied covariance is non-positive-definite (minimum eigenvalue -0.35 at 0.45) and the contaminated DGP is no longer faithfully representable.

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The production run (1{,}000 replicates per cell, balanced-placebo design with three open-label, six covert on-drug, and six open-placebo timepoints, $N \in \{70, 150\}$, $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$) confirms the recovery at full resolution. At $N = 150$ the mean estimated $\hat{\beta}_{bm:D_{bc}}$ from the decomposition tracks the pharmacological target -2.25 across the entire contamination range, with bias -0.022, $+0.008$, $+0.021$, and -0.026 at $c_{bm,PB} = 0, 0.1, 0.2, 0.3$ respectively, every value within one Monte Carlo standard error (about 0.02) of zero. The one-component estimator on the same data is unbiased only at $c_{bm,PB} = 0$ and develops a monotone bias of $+0.16$, $+0.34$, and $+0.48$ as the biomarker-PB correlation grows. Calibration tracks the bias: the decomposition holds near-nominal interval coverage (0.94 to 0.96), whereas one-component coverage degrades to 0.86 at $c_{bm,PB} = 0.3$. The $N = 70$ cells show the same pattern with the decomposition unbiased (absolute bias ≤ 0.08) and lower power (about 0.66 versus 0.93 at $N = 150$), the precision cost of the smaller sample and the extra belief-covariate terms. Convergence was complete in every cell. Per-replicate output and the cell-level summary are at `analysis/data/derived_data/component-decomposition/study-b-balanced/`. The contamination range is bounded above by positive-definiteness: at $c_{bm,PB} = 0.45$ the biomarker would have to correlate 0.45 with both BR and PB, which the covariance cannot accommodate (minimum eigenvalue -0.35), so we restrict the production grid to the feasible range $c_{bm,PB} \leq 0.30$.

[87]

Recovery is therefore conditional on the design, not on the analysis alone: the decomposition is necessary but the belief-decoupling design is what makes the pharmacological slope identifiable.

- The practical implication is that a biomarker-validation trial exposed to possible biomarker-PB coupling should adopt a balanced-placebo or open-hidden design, not the standard hybrid design, if the pharmacological interaction is to be recovered.
- This sharpens the framework's central claim: decomposition is

the mechanism, but a design that varies belief independently of drug exposure is the enabling condition.

- Open questions remaining: recovery beyond the positive-definite range, the participant-level component recovery of the original Study B identifiability aims, and replication of the pattern under Architecture A (mean-moderation) coupling.

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The result sharpens the paper's central claim in an important way. The inferential failure of section 7.4 is real and the decomposition does remove it, but the decomposition is not self-sufficient: on the standard hybrid design it does not recover the pharmacological slope, because the design does not supply independent variation in drug exposure and belief. The enabling condition is a design that decouples the two, of which the balanced-placebo and open-hidden paradigms[51,52] are the established instances. The practical recommendation that follows is specific: a biomarker-validation trial in which the candidate biomarker may correlate with placebo responsiveness should be fielded as a balanced-placebo or open-hidden design rather than the standard hybrid design, because only the former makes the pharmacological biomarker-by-treatment interaction separately identifiable. Three questions remain open. Recovery beyond the positive-definite contamination range is not characterised, since the MVN architecture cannot represent it; the participant-level component recovery that motivated the original identifiability aims of Study B is not addressed by the slope-level analysis here; and whether the same recovery holds under Architecture A (mean-moderation) coupling, where the biomarker shifts component means rather than correlating with component draws, remains to be confirmed.

8 Application to predictive-biomarker validation

[88]

The precision-medicine question – whether a biomarker predicts pharmacological response – is answered by the lumped total only when the biomarker is uncorrelated with PB and TV; otherwise it requires decomposing the response.

- The question is whether the biomarker predicts the BR component, not the lumped total.
- The PATH statement formalises this as the target estimand for predictive biomarker validation.
- The BR-PB-TV decomposition is the mechanism that separates the pharmacological signal when biomarker-component coupling

cannot be ruled out.

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The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalised in the PATH statement[14]. Whether it depends on the BR-PB-TV decomposition is itself conditional: by the identity of section 6.1, the lumped biomarker slope equals the pharmacological slope when the biomarker is uncorrelated with PB and TV, and departs from it only by the biomarker's covariance with those components. The decomposition is therefore the mechanism the question requires precisely in the regime where that covariance is non-negligible.

[89]

A lumped-total biomarker regression conflates pharmacological and placebo-belief prediction, producing a stratification biomarker that mis-identifies responders.

- $\hat{\beta}_{bm}$ from the lumped regression contains both pharmacological and placebo-belief prediction in unknown proportions.
- If PB covaries with the biomarker distribution, flagged 'high responders' include high-PB responders who will not benefit pharmacologically.
- Prospective application of such a biomarker would systematically mis-stratify patients for prescription decisions.

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A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the lumped total contains. If PB varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then $\hat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

[90]

A valid pharmacological predictor requires a non-zero $\hat{\beta}_{bm}^{BR}$; significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ indicate a psychological or prognostic biomarker, not a pharmacological one.

- The sole criterion for pharmacological prediction is a meaningfully non-zero BR-component regression coefficient.
- A biomarker correlating only with PB selects placebo responders, not drug responders.
- A biomarker correlating only with TV is prognostic but not predictive in the pharmacological sense.

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The right approach is to regress each component on the biomarker:

$$\begin{aligned} BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\ PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\ TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}. \end{aligned}$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

[91]

When biomarker-component coupling cannot be ruled out, the decomposition is the mechanism by which the precision-medicine question is posed; when it can, the lumped analysis already answers it.

- Where the biomarker may correlate with PB or TV, separating BR from them is what prevents validation from conflating pharmacological and non-pharmacological prediction.
- The phase-augmented `bm:Dbc:phase` interaction is a small-N proxy, but Study A shows it buys nothing under an uncontaminated biomarker, so it is warranted only when contamination is plausible.
- The full decomposition recovers component-specific regressions explicitly when sample size permits.

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The decomposition is therefore the mechanism by which the precision-medicine question is posed whenever the candidate biomarker may correlate with PB or TV. Where the biomarker can be assumed orthogonal to the non-pharmacological components, the identity of section 6.1 shows the lumped analysis already targets the biomarker-by-BR estimand, and the simulation of section 7.3 confirms it recovers that estimand without bias. The phase-augmented linear-mixed analysis introduced in section 5 retains a useful subset of the decomposition machinery at small N (the `bm:Dbc:phase` interaction tests whether the biomarker-by-treatment effect attenuates under blinding, the practical proxy for biomarker-by-PB contamination), but Study A shows this proxy earns its degrees of freedom only when such contamination is present: under an uncontaminated biomarker it reduced no bias and lost power. The full decomposition recovers the component-specific regressions explicitly when N permits and when the scientific question warrants the cost.

9 Discussion

[92]

The three-component framework reframes treatment response as a structured sum of causally distinct contributions, and the present work maps when acting on that decomposition actually changes inference.

- The framework specifies which trial-design contrasts identify which components.
- Acting on the decomposition is valuable conditionally: for attribution estimands, and for biomarkers that may correlate with PB or TV, but not for a biomarker-by-treatment interaction with a biomarker orthogonal to them.
- The target estimand for biomarker validation is the biomarker-by-BR association; whether a lumped analysis already recovers it depends on the biomarker's coupling to PB and TV.

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The three-component decomposition is the formal expression of the claim that 'treatment response' in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. What the present work establishes, analytically in section 6.1 and empirically in section 7.3, is that the inferential value of acting on this decomposition is conditional rather than universal. The identity shows that the one-component estimator is displaced from its pharmacological target only by the components' loading onto the contrast used, and, for a biomarker slope, only by the biomarker's covariance with PB and TV. Where those terms are zero, the design-contrast one-component analysis is already unbiased, and the production simulation confirms it: with a biomarker coupled to BR alone, the one-component biomarker-by-treatment estimate is unbiased across the whole PB-TV grid and the phase-augmented analysis is dominated. The framework's contribution is therefore best read as a map of when decomposition changes inference, not as a blanket prescription to decompose. It makes the trial-design conditions explicit, identifies the attribution and contaminated-biomarker estimands for which decomposition is needed, and clarifies the target estimand for biomarker validation (the biomarker-by-BR association, not the lumped total).

[93]

Four open questions bound the framework's current scope: the simulation results are pending, the η multipliers require sensitivity analysis, non-monotone natural history is not yet handled, and participant-level recovery inherits the identifiability caveats of individual treatment effects.

- Quantitative bias-versus-power trade-offs are conjectures until the section 7 simulation is run.
- The η(phase) expectancy-modulation parameters are modelling assumptions requiring explicit sensitivity analysis.
- Cyclical or biphasic TV trajectories exceed the Gompertz form and require flexible or covariate-augmented specifications.
- Participant-level component recovery rests on separating participant-by-component signal from participant-by-period and participant-by-phase variation, the identifiability caveat Senn raises for individual treatment effects.

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Four open questions remain and warrant acknowledgement. First, the simulation study described in section 7 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the η(phase) multipliers used to parameterise the expectancy modulation of PB are themselves modelling assumptions, and a sensitivity analysis varying η across plausible values is mandatory before any clinical conclusion is drawn from the framework's outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic TV patterns require either flexible TV specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that[54] and[49] raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

[94]

Two companion documents extend the manuscript: a pedagogical narrative and a Gompertz-family sensitivity evaluation.

- The pedagogy document at `docs/24-component-decomposition-pedagogy.md` provides worked examples for first-time readers.
- The Gompertz evaluation at `analysis/report/07-gompertz-evaluation/` quantifies trajectory-specification sensitivity.
- Together they address the intuitive and parametric dimensions of the framework that the present manuscript treats briefly.

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The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

[95]

The components exist in any longitudinal treatment response, but participant-level identifiability requires a within-subject belief-by-exposure contrast that not every placebo-controlled design supplies.

- The three-component structure is a property of the data-generating process of any repeated-measures treatment study; its identifiability is design-specific.

- Parallel-group placebo control identifies the population-mean PB but not a participant-level PB, and supplies no within-subject belief contrast; only designs with such a contrast (blinded or open-hidden discontinuation, randomised withdrawal, crossover, hybrid N-of-1) identify the participant-level decomposition the biomarker question needs.
- The broad claim is therefore scoped to designs supplying a within-subject belief-by-exposure contrast, not to placebo-controlled longitudinal trials in general.

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A final point concerns the scope of the decomposition, where it is essential to distinguish the existence of the components from their identifiability. The claim that an observed longitudinal treatment response is the sum of a pharmacological, a belief-driven, and a natural-history component is a statement about the data-generating process of any repeated-measures treatment study, and in that sense the decomposition is not specific to N-of-1 trials. Identifiability, by contrast, is design-specific, and the distinction between population-level and participant-level recovery is decisive for the biomarker application. A standard parallel-group placebo-controlled trial supplies a between-arm placebo contrast that identifies the population-mean belief response but not a participant-level PB, and it contains no within-subject contrast in which belief varies independently of pharmacology; the participant-level biomarker-by-BR slope that the precision-medicine question targets is therefore not identified from such a design, no matter how large it is. The designs that do supply the required within-subject belief contrast are the blinded or open-hidden discontinuation and randomised-withdrawal designs[51,52], the crossover with on/off periods, and the hybrid N-of-1 design developed here; the pharmacometric disease-progression-plus-placebo models from which the framework descends[1–3] operate at the group level and recover population-mean components only. We therefore advocate the decomposition as a default lens for longitudinal designs that supply a within-subject belief-by-exposure contrast, with the aggregated N-of-1 case the most demanding instance because it must recover the components from a single participant's trajectory. We do not claim participant-level identifiability from parallel-group placebo control alone, and where only a between-arm placebo contrast is available the framework's claims are restricted to the population mean.

10 Conclusions

[96]

Treatment response in aggregated N-of-1 trials is the sum of three causally distinct components, and the bias from conflating them is governed by the estimand and the biomarker's coupling to the components, not by their raw magnitude.

- The three components are BR (pharmacological), PB (placebo-belief), and TV (natural history).
- Each has its own biological mechanism, identifiability conditions, and clinical interpretation.
- The lumped pre-post treatment effect is biased by the PB and TV contributions; the biomarker-by-treatment slope is biased only by the biomarker's covariance with PB and TV, as Study A confirms.

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In conclusion, three points are to be emphasised. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR), placebo-belief response (PB), and time-variant natural history (TV). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation. Whether conflating them biases a given analysis is, however, governed by the omitted-variable-bias identity of section 6.1, not by the raw magnitude of PB and TV: the lumped pre-post treatment effect is displaced by the PB and TV contributions, whereas the biomarker-by-treatment slope is displaced only by the biomarker's covariance with those components. Study A bears this out: varying PB and TV magnitude with a biomarker coupled to BR alone left the one-component estimate unbiased across the entire grid.

[97]

What matters is the design contrast, not the parametric model: the trial must supply the drug-on/off and belief contrasts, but adding parametric component terms is not free and is not always worth it.

- Unique component identification requires phase contrasts built into the trial design; the one-component analysis already exploits the drug-on/off contrast.
- Under an uncontaminated biomarker, the phase-augmented analysis adds no bias reduction and loses power, so it is not recommended as a default.
- The phase-augmented and full component-matched analyses are warranted for attribution estimands and for biomarkers that may

correlate with PB or TV, where the extra structure does work.

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Second, the operative requirement is the design contrast, not the parametric analysis model. The trial design must include the drug-on/off and belief contrasts that identify the components, and the one-component analysis already exploits the drug-on/off contrast through its D_{bc} term. Adding parametric component structure on top is not free: under an uncontaminated biomarker the phase-augmented analysis reduced no bias and lost substantial power, so it should not be adopted as a default. Its value, and that of the full component-matched fit, is confined to the estimands where the extra structure does work, namely the attribution of response to pharmacology versus belief and the analysis of biomarkers that may themselves correlate with PB or TV.

[98]

For predictive-biomarker validation the decomposition matters precisely when the candidate biomarker may correlate with PB or TV; the contaminated-biomarker study demonstrates this failure and Study B shows it is recoverable under a belief-decoupling design.

- A biomarker that predicts the lumped total is a pharmacological predictor only if it is uncorrelated with PB and TV; otherwise the lumped slope is contaminated.
- Regulatory and prescribing decisions therefore require separating BR from PB and TV whenever biomarker-component coupling cannot be ruled out.
- Study A established the uncontaminated baseline, the contaminated-biomarker study (section 7.4) demonstrated the failure, and Study B (section 7.5) showed the bias is recoverable on a balanced-placebo design but not on the standard hybrid design.

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Third, for predictive-biomarker validation the decomposition earns its place precisely when the candidate biomarker may correlate with PB or TV. A biomarker that predicts the lumped total is a pharmacological predictor only when it is uncorrelated with the non-pharmacological components; where that cannot be assumed, the lumped slope mixes pharmacological and non-pharmacological prediction, and the regulatory and prescribing decisions that motivate the trial require the separation the decomposition supplies. The contaminated-biomarker study (section 7.4) demonstrates this failure directly, and Study B (section 7.5) establishes that the bias is recoverable: a belief-covariate decomposition returns the unbiased pharmacological slope across the feasible contamination range, but only on a balanced-placebo design that decouples drug exposure from belief, not on the standard hybrid design. The decisive practical lesson is therefore as much about design as about analysis.

11 Conflict of interest

The authors declare no conflicts of interest.

12 Funding

This work received no specific funding.

13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

14 Reproducibility

[99]

The manuscript is rendered inside a Docker container with locked package versions; the full software environment is pinned by `Dockerfile` and `renv.lock`.

- R 4.4.x in the zzcollab Docker container provides the execution environment.
- Simulation packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.
- The manuscript build uses `rmarkdown` and `bookdown`.

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This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[100]

Reproducibility is ensured by a master seed of 20260507 with per-replicate seeds derived by index addition.

- `set.seed(20260507)` is called at the top of each replicate loop.
- `parallel::mc.set.seed` propagates seeds inside `mclapply`.
- Per-replicate seeds are `master_seed + rep_idx`.

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Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

[101]

All simulation outputs, driver scripts, pre-registration, and manuscript source files are versioned at documented paths within the repository.

- Per-replicate outputs and Morris summaries are at `analysis/data/quick-sim/`.
- Driver scripts and the ADEMP pre-registration are at `analysis/scripts/component-decomposition/`.
- Manuscript source, bibliography, and CSL file are at `analysis/report/06-component-decomposition/`.

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Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates`
Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`.
Driver scripts are under `analysis/scripts/component-decomposition/`.
The pre-registered ADEMP plan is at `analysis/scripts/component-decomposition/00-ademp-pre-reg`
The manuscript source, paper-specific bibliography (`references.bib`),
and SIM CSL file are at `analysis/report/06-component-decomposition/`.

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